Tribhuvan University
Faculty of Humanities and Social Science

A Project Report on
Bus Reservation System

Submitted to
Department of Computer Application
Nepalaya Education Foundation

In Partial fulfilment of the requirements of Bachelors in Computer Application

Submitted By
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SUPERVISOR'S RECOMMENDATION

I hereby recommend that this project proposal under my supervision by **Udesh Regmi & Sangam Sedai** entitled **"Web Based Bus Reservation System"** in partial fulfilment of the requirements for the degree of Bachelor in Computer Application is recommended for the final evaluation.

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SIGNATURE

Yubaraj Devkota

(SUPERVISOR)

Bachelor in Computer Application

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LETTER OF APPROVAL

This is to certify that this project proposed by **"Udesh Regmi and Sangam Sedai"** entitled **"Web Based Bus Reservation System"** in partial fulfilment of the requirements for the degree Bachelor in Computer Application has been evaluated. In our opinion it is satisfactory in the scope and quality as a project for the required degree.

Yubaraj Devkota, Supervisor Nepalaya Education Foundation	Kapil Bam Thakuri, Coordinator Nepalaya Education Foundation
Internal Examiner	External Examiner

ABSTRACT

The rapid advancement in information technology has revolutionized the transportation sector, particularly in making travel more accessible and efficient. This project presents the development of an Online Based Bus Reservation System (OBRS), a web-based application designed to streamline the process of booking bus tickets for passengers while providing administrative tools for bus operators. Traditional bus reservation methods often involve long queues, manual record-keeping, and limited availability, leading to inefficiencies and customer dissatisfaction. OBRS addresses these challenges by offering a user-friendly online platform where passengers can search for available buses, view routes and schedules, select seats, make secure payments, and receive e-tickets instantly. The system is built using PHP as the backend scripting language, MySQL for database management, and HTML/CSS/JavaScript with Bootstrap for the frontend interface, ensuring responsiveness across devices. For administrators, the system includes modules for managing bus details, user accounts, bookings, and generating reports. The implementation follows a structured methodology, incorporating feasibility studies, entity-relationship modeling, and data flow diagrams to ensure robustness. Through this system, the project aims to reduce booking time by 70%, enhance data accuracy, and improve overall user satisfaction. Extensive testing validates the system's reliability, security, and scalability, making it suitable for small to medium-scale bus service providers. This work contributes to the digital transformation of public transport, promoting paperless operations and real-time updates.

Keywords: Online Bus Reservation, Web Application, PHP MySQL, Seat Booking, Admin Panel, E-Ticket Generation, Transportation Management

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LIST OF ABBRIVATIONS

API: Application Programming Interface

CSS: Cascading Style Sheets

DFD: Data Flow Diagram

ER: Entity-Relationship

HTML: HyperText Markup Language

JS: JavaScript

MySQL: My Structured Query Language (Database Management System)

OBRS: Online Based Bus Reservation System

PHP: Hypertext Preprocessor (Server-Side Scripting Language)

SQL: Structured Query Language

UI: User Interface

UX: User Experience

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Chapter 1: Introduction

1.1 Introduction

An Online Bus Reservation System is a web-based platform designed to automate and simplify the process of booking bus tickets, managing reservations, and handling passenger information. The system enables passengers to book seats online in real time, reducing the need for physical reservations and long queues at ticket counters. With the increasing demand for digital services, people now expect fast, secure, and reliable solutions for travel planning. A bus reservation system provides a convenient interface where users can search for routes, view bus schedules, check seat availability, and complete bookings with online payment via esewa or cash on counter. Administrators can manage bus details, reservations, seat allocations, and generate reports efficiently. By offering an easy-to-use and accessible platform, the Online Bus Reservation System enhances customer satisfaction, reduces manual workload, and improves operational efficiency for transport companies.

1.2 Problem Statement

In the contemporary era of digitalization, the transportation industry, particularly bus services, continues to rely heavily on outdated manual processes for ticket reservations. Passengers often face significant challenges such as long waiting lines at bus terminals, inconsistent availability of seats, errors in manual booking records, and lack of real-time information on bus schedules and routes. These inefficiencies not only result in time wastage but also lead to overbooking, revenue loss for operators, and frustration among users who may miss out on preferred travel options. Moreover, during peak seasons like holidays or festivals, the system becomes overwhelmed, exacerbating delays and inaccuracies. Security concerns, such as fake tickets and data privacy issues, further compound the problems. Traditional methods also hinder scalability, as expanding services requires proportional increases in physical infrastructure and staff, which is neither cost-effective nor environmentally friendly. This project addresses these pain points by developing an online platform that automates the entire reservation workflow, ensuring seamless access, accurate data management, and enhanced user satisfaction. By leveraging web technologies, the system eliminates geographical barriers, allowing users from remote areas to book tickets effortlessly via the internet. Ultimately, the descriptive analysis reveals that without such a digital intervention, the bus reservation sector risks stagnation in a rapidly evolving technological landscape, impacting both service providers and commuters adversely.

1.3 Objectives

- To develop a robust web application utilizing PHP for server-side logic, MySQL for relational database management, HTML/CSS/JavaScript for responsive frontend design, ensuring secure and efficient handling of user interactions and data persistence.
- The system shall enable passengers to register, search and book bus seats, process payments, and download e-tickets; while administrators can manage bus routes, monitor bookings, handle user queries, thereby facilitating end-to-end reservation management.

1.4 Scope and Limitations

Scope: The OBRS encompasses a comprehensive web platform accessible via browsers on desktops and mobiles, supporting user registration, secure login, real-time bus search by route, seat selection with availability checks, online payment integration (via simulated gateway), e-ticket generation, booking history tracking, and cancellation. For administrators, it includes dashboard for adding/editing bus details, user management, booking oversight, revenue analytics, and system backups. The system supports multiple bus operators, up to 100 daily routes, and scales for 1,000 concurrent users. Future extensions can integrate SMS alerts and mobile apps. It promotes paperless operations, data analytics for demand forecasting, and multilingual support for broader accessibility.

Limitations: Currently, payment is simulated (no live gateway integration due to project constraints); tracking of buses while moving across the routes are not implemented; the system handles only domestic routes within one country; high-traffic scalability testing is limited to simulated loads; and dependency on stable internet connectivity may affect offline users in remote areas.

1.5 Report Organization

This report is organized into several structured chapters to provide a clear understanding of the Online Bus Reservation System its purpose, design, implementation, and overall functionality. Each chapter focuses on a specific aspect of the system so that readers can grasp both the technical and conceptual details effectively.

- **Chapter 1: Introduction**
  This chapter provides an overview of the Online Bus Reservation System, including the background of the tours and travelling industry specially bus & the motivation behind developing the system, and the problems it aims to solve. It explains the scope, objectives, and significance of the project.
- **Chapter 2: Literature Review / Background Study**
  This section presents an analysis of existing reservation systems, booking platforms, and bus portals. It identifies their limitations and highlights the areas where OBRS, improves upon current solutions, particularly in user convenience, automation, and integrated travel services.
- **Chapter 3: System Analysis and Design**
  This chapter includes requirement analysis, feasibility study, and problem definition. Both functional and non-functional requirements are documented, along with use-case descriptions that illustrate how users interact with the system. System workflows and design strategies for the Online Bus Reservation platform are also discussed.
- **Chapter 4: Implementation and Testing**
  This chapter explains the system architecture, database schema, Data Flow Diagrams (DFDs), Entity-Relationship Diagrams (ERDs), and the overall development methodology. It describes how different components—such as bus reservation, bus adding and update and how data flows throughout the Online Bus Reservation System. Testing procedures and results are also included.
- **Chapter 5: Conclusion and Future Recommendations**
  This section summarizes the technologies used, the development tools, coding structure, and the module-wise implementation of the Online Bus Reservation System. Screenshots and interface descriptions are provided to demonstrate system functionality. The chapter concludes with recommendations for future enhancements and additional features.

Chapter 2: Background Study and Literature Review

1.1 Background Study

Nepal’s public transportation system has undergone a significant transformation, evolving from rudimentary manual operations to increasingly sophisticated digital platforms. This evolution has been shaped by the country’s diverse topography, ranging from mountainous terrains to plains, and by socio-economic factors that influence commuter needs and transport accessibility. Historically, public transportation relied heavily on cooperative and community-based initiatives, with Sajha Yatayat, established in 1961, playing a pivotal role in providing affordable and reliable mobility services. Initially focused on cooperative bus operations to serve urban and inter-provincial travelers, Sajha Yatayat has progressively integrated modern elements such as electric vehicles, basic online portals for route and schedule information, and limited digital booking functionalities. These efforts reflect an ongoing attempt to balance affordability, sustainability, and technological adoption in a market characterized by varied demand and operational constraints.

In the context of system development for Nepal’s transport sector, foundational concepts such as CRUD (Create, Read, Update, Delete) operations, entity-relationship modeling, and data flow diagrams are essential for designing efficient, scalable, and maintainable web-based platforms. CRUD operations facilitate systematic management of reservation and passenger data, while ER modeling ensures coherent database structures, and DFDs allow developers to optimize operational processes, reducing redundancies and enhancing workflow efficiency. Although many operators have traditionally relied on desktop-based applications for internal record-keeping, such tools often lack accessibility for end-users and real-time connectivity, limiting their usefulness in a modern digital ecosystem. The increasing penetration of mobile internet in Nepal highlights the importance of web-based platforms that enable remote booking, real-time updates, and broader accessibility for both commuters and transport operators.

Methodologically, frameworks like the Waterfall model provide a structured approach to system development, particularly when sequential phases such as requirement analysis, design, implementation, testing, and deployment are necessary. Coupled with feasibility studies tailored to Nepal’s economic context, these approaches ensure that solutions remain cost-effective and practically implementable, often leveraging open-source tools to mitigate financial barriers. Such rigorous planning and design are crucial in a setting where infrastructural challenges, connectivity limitations, and operator diversity present significant hurdles to effective service delivery.

Within this landscape, the Online Based Bus Reservation System (OBRS) emerges as a comprehensive solution aimed at addressing both operational inefficiencies and accessibility gaps. By integrating dual client–admin functionalities, OBRS enables end-users to perform seamless bookings while empowering operators with real-time management, reporting, and control over schedules and inventory. The system's design emphasizes scalability, reliability, and inclusivity, ensuring that even smaller or remote transport operators can participate effectively in Nepal's digital transport ecosystem. Furthermore, OBRS aligns with global best practices in transportation digitization, incorporating key features such as user-friendly interfaces, secure payment gateways, and structured database management to improve both passenger experience and operator efficiency.

Overall, the historical evolution of Nepal's public transportation, coupled with the current push toward digitalization, underscores the pressing need for integrated, web-based platforms like OBRS. By bridging technological gaps, enhancing administrative capabilities, and fostering equitable access to transport services, OBRS represents a critical advancement in modernizing Nepal's bus reservation systems and supporting sustainable, reliable mobility across the country.

1.2 Literature Review

The rise of digital technologies has transformed Nepal's transportation sector, allowing commuters and tourists to access bus services online despite challenging geography. Various platforms now support route searches, seat selection, payments, and confirmations. Understanding these systems is essential for developing advanced Bus Reservation Systems that address current gaps and improve user and operator experiences. This section critically reviews major Nepali bus reservation platforms and related research, assessing their strengths, limitations, and relevance to OBRS, a comprehensive system tailored to Nepal's needs.

2.2.1 Bus Sewa

BusSewa, developed by Diyalo Technologies Pvt. Ltd., is a major online bus reservation platform in Nepal, covering over 300 routes across 65 districts and partnering with more than 300 operators, including Sajha Yatayat. The platform enables users to search routes, compare schedules, check seat availability, and book tickets online, with secure payments via eSewa and Khalti, often including promotional discounts. Its 24/7, user-friendly design simplifies travel for commuters and tourists. However, BusSewa faces limitations such as performance issues during peak traffic, limited administrative control for operators, and lack of support for cross-border routes or integrated travel services. Its commission model may also hinder smaller operators. These gaps underscore the need for scalable systems like OBRS, offering comprehensive admin and CRUD functionalities to improve inclusivity in Nepal's transport sector [1]

2.2.2 West Nepal Bus (Paschim Nepal Bus Sewa)

West Nepal Bus, operated by Paschim Nepal Bus Byabasai Pvt. Ltd., offers online reservations for western Nepal routes, including Kathmandu to Lumbini and Butwal, with options for AC Deluxe and standard buses and direct payments. Launched around 2017, it provides search by date, shift, and destination, along with instant confirmations, serving regional travelers in provinces like Gandaki and Lumbini. Its localized focus supports reliable service, promotes tourism, and reduces overbooking through basic real-time updates. However, route coverage is limited, there is no mobile app, and administrative tools are minimal. Users needing eastern or far-western connections face fragmented experiences, and the lack of multilingual support may affect tourists. These gaps point to opportunities for OBRS to provide national coverage with flexible admin features, improving efficiency in Nepal's decentralized bus network [2].

2.2.3 Bus Sewa Nepal

Bus Sewa Nepal, run by Small Heaven Travel and Tours Pvt. Ltd., provides online ticketing with transparent pricing, cashback incentives, and detailed bus information, including seat capacity and e-ticketing. Its user-friendly interface and verified operators foster trust, supporting cost-conscious travellers and tourists. However, route coverage is limited, bookings rely on email confirmations, and administrative tools for real-time schedule and bulk management are lacking. Restricted payment options also pose challenges for international users. These limitations underscore the need for OBRS, which integrates client–admin modules to improve efficiency and reliability [3].

2.2.4 Sajha Yatayat

Sajha Yatayat, founded in 1961, offers digital services via its website and app for route, schedule, and fare information, focusing on affordable and eco-friendly transport in the Kathmandu Valley and select inter-provincial routes. Its community-oriented model benefits low-income commuters and promotes sustainable mobility. However, the system lacks full online booking and payment options, and administrative tools for real-time fleet management are limited, reducing efficiency. Research on Nepal's public transport digitization highlights these gaps, emphasizing the need for systems like OBRS with comprehensive CRUD modules to modernize operations [4].

2.2.5 Purba Mechi Bus (Purba Nepal Bus Sewa)

Purba Mechi Bus Pvt. Ltd. offers online booking for eastern Nepal routes with free reservations, online payments, and international visibility through platforms like 12Go. It supports regional commuters and tourism by improving access to underserved eastern corridors. However, limited real-time seat mapping and weak admin controls reduce operational flexibility. Studies on eastern Nepal's bus networks note similar inefficiencies, underscoring the need for integrated systems like OBRS, which improves consistency through structured ER and DFD modeling [5].

2.2.6 Panch Pokhari Sofa Seater

Panch Pokhari Sofa Seater provides premium bus services on routes from Kathmandu to eastern Nepal, offering enhanced comfort for long-distance travelers. Its focus on passenger amenities improves travel experience in a competitive market. However, the service relies on third-party platforms for bookings and lacks independent admin and CRUD capabilities, leading to potential data inconsistencies. Research on Nepal's reservation systems highlights these issues, supporting the need for unified platforms like OBRS to enable seamless integration and scalability [6].

2.2.7 Bus Tracking and Reservation System

Malik et al. propose a MERN-based Bus Tracking and Reservation System with real-time GPS tracking, online booking, secure payments, and route optimization. The system reduces wait times through delay notifications and a centralized database. While not Nepal-specific, its real-time and mobile features are relevant for developing urban transport systems. Limitations include legacy system integration and reliance on stable internet. These insights support OBRS's use of CRUD features to improve reliability in Nepal's bus networks [7].

2.2.8 Online Travel Portals and Bus Ticket Reservation System

Joshi et al. (December 2021) examine online travel portals and propose a Bus Ticket Reservation System to address Nepal's manual scheduling, ticketing, and revenue-tracking inefficiencies. The system includes user-friendly interfaces for bookings, cancellations, and refunds, aiming to reduce errors and conflicts. Its strength lies in targeting Nepal-specific needs to improve operational efficiency. Limitations involve payment and refund concerns, system vulnerabilities, and connectivity issues. The study highlights the importance of customer-focused design, reinforcing OBRS's approach of secure, integrated client–admin modules to streamline Nepal's transport sector [8].

2.2.9 Nepal as a Destination for Foreign Female Travelers

Karki's 2023 thesis explores transportation safety for female solo travelers in Nepal, highlighting both public and tourist buses. While tourist buses provide air-conditioned comfort and seat reservations, bookings often require in-person visits, and public buses face issues like overcrowding and harassment. Buses remain affordable and widely available, but poor roads, accidents, and limited online services are major drawbacks. The study emphasizes digital apps for safer, more transparent travel, indicating opportunities for OBRS to offer web-based booking and safety features. Recommendations such as insurance and safety awareness align with OBRS's potential to improve accessibility for vulnerable users [9].

Chapter 3: System Analysis and Design

3.1 System Analysis

System analysis is the process of studying and understanding a system so it can be developed in an organized way. For OBRS, the requirements are clear and stable, so the project follows the waterfall model. Each phase is completed before moving to the next, which helps maintain a proper flow and ensures the system is built in a systematic and reliable manner. This method develops the system step-by-step through the following phases:

- Requirement
- Design
- Implementation
- Testing
- Documentation

In Waterfall, every phase relies upon the results of the previous phase. Prior to moving and going for the next phase, a proper review of the previous phase is conducted, and this helps keep everything consistent and clear in the whole process of development. As this proposed system has fixed specifications and does not need many adjustments and additions, the Waterfall model is the best approach for developing software related to the OBRS.

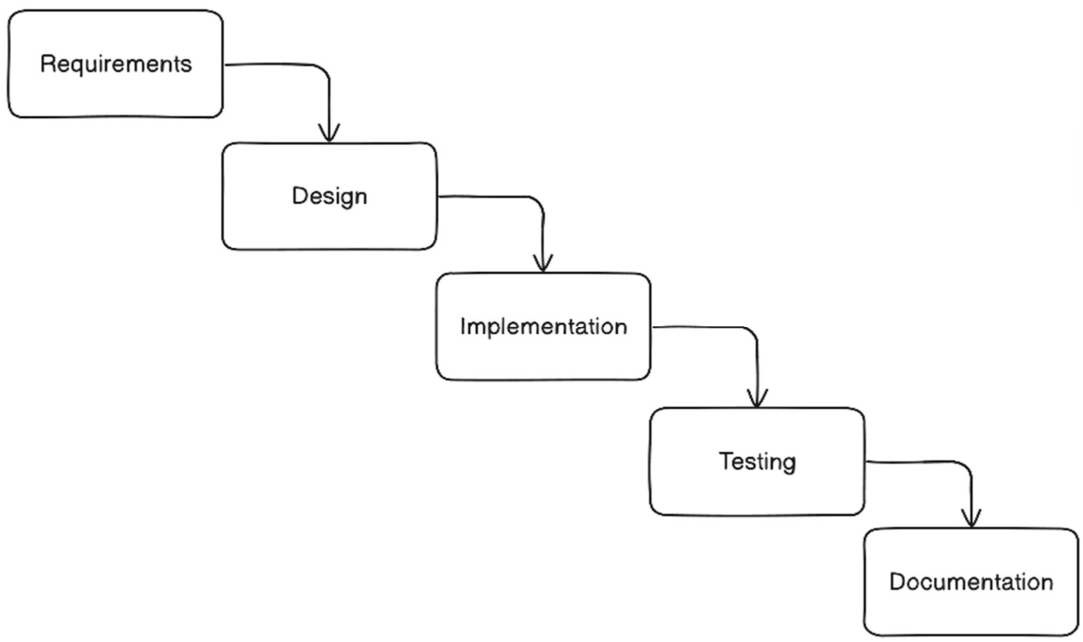


Figure 1 Waterfall Methodology of OBRS

3.1.1 Requirement Analysis

The system requires the following functional and non-functional requirements

- **Functional Requirement:**
    i. User Registration and Authentication.
    ii. Reservation Management
    iii. Admin Management

  A use case is a methodology used in system analysis to identify, clarify and organize system requirements. Use case diagram for the proposed system is given below:

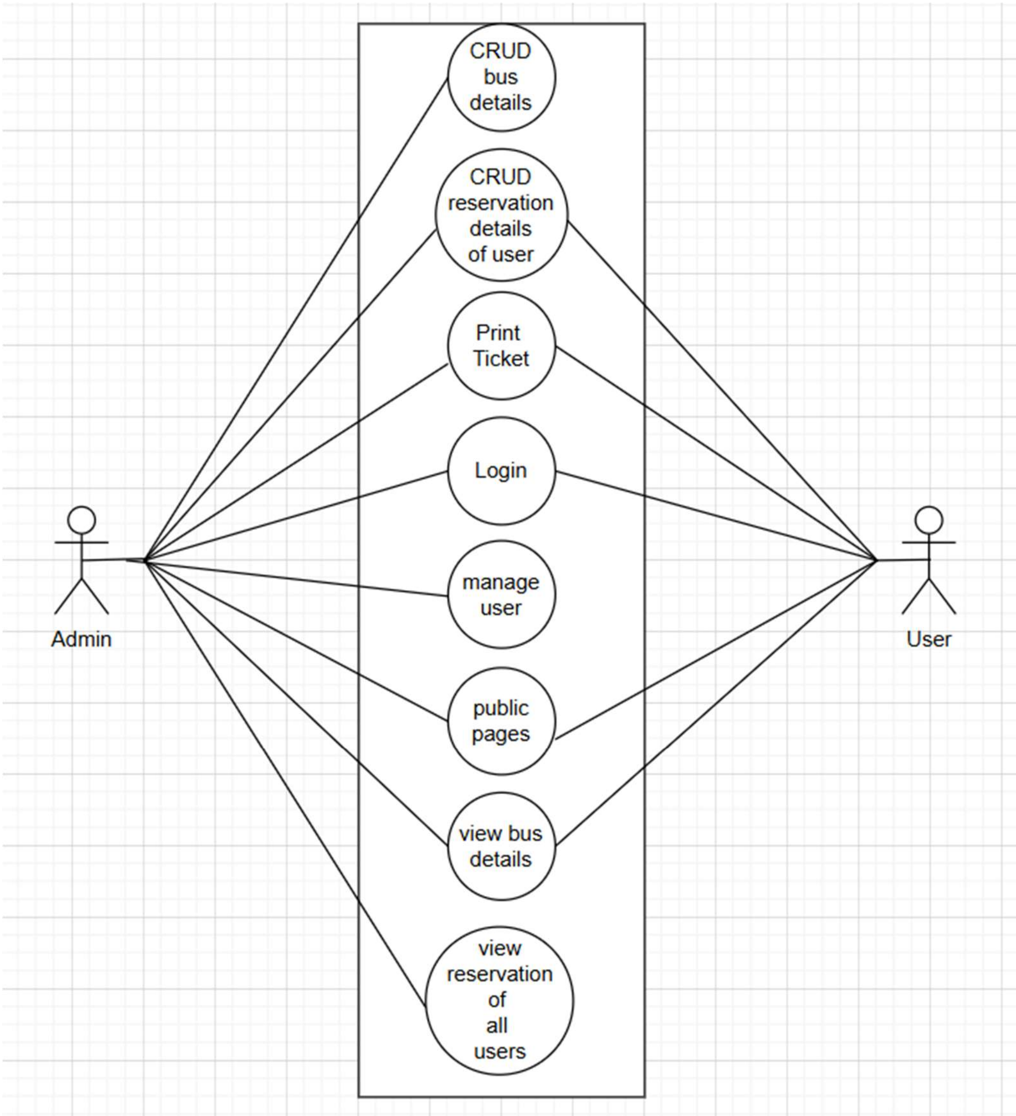


Figure 2. Use Case diagram of OBRS

- **Non-Functional Requirement:**
  i. **Security:** only authorized person can have access to this system.
  ii. **Performance:** This system is designed for smooth performance with good optimization and good response.
  iii. **Reliability:** This system will be reliable for the users, the system will run 24/7.
  iv. **User Friendly:** This system consists of user friendly and interactive

3.1.2 Feasibility Analysis:

A feasibility study assesses the practicality of the proposed Online Based Bus Reservation System (OBRS) by evaluating technical, operational, economic, and schedule aspects prior to development.

- **Technical Feasibility:**

The system is technically viable using free, open-source technologies: HTML, CSS, and JavaScript for the frontend, PHP for server-side scripting, and MySQL for database management. These tools support full CRUD operations without external APIs.

Table 1 Technical Feasibility Study

Technological Requirement	Hardware Requirement	Software Requirement
HTML	Laptop / PC	VS Code
CSS	Minimum 4GB RAM	Figma
JS	Keyboard	XAMPP Server
MYSQL	Mouse	MySQL /phpMyAdmin
PHP		MS Word
		Web Browser

- **Operational Feasibility**:

The system is operationally feasible, designed for users with varying digital literacy in Nepal. Passengers can search routes, select seats, and generate e-tickets through an intuitive web interface, while administrators manage buses, routes, and bookings via a secure dashboard. Minimal training is needed, with a short manual for users and basic guidance for admins on CRUD operations. Its web-based design allows deployment on affordable local hosting, ensuring easy maintenance and updates. High user acceptance is expected due to the growing use of online ticketing among Nepali commuters and tourists.

- **Economic Feasibility**:

OBRS is economically viable, relying on free open-source tools with minimal costs for domain registration (NPR 1,000–2,000/year) and shared hosting (NPR 3,000–8,000/year). Hardware requirements are met by existing laptops, and maintenance costs remain low, scaling with user demand. The system reduces manual labor, prevents overbooking, and enables revenue from premium features. ROI is quick, especially for medium-sized operators, making OBRS highly cost-effective in Nepal.

- **Schedule Feasibility:**

Schedule feasibility is critical for project success, as delays could render the system outdated. The project timeline spans four months, following the Waterfall methodology with defined phases: requirements gathering (1 week), design (3 weeks), implementation (8 weeks), testing (3 weeks), and documentation/deployment (2 weeks).

A Gantt chart was prepared to visualize and monitor progress, ensuring milestones are met within the BCA semester constraints. Tasks were allocated considering the team's capacity (two members), with buffers for unforeseen issues like testing iterations. This structured approach confirms that the project can be completed on time.

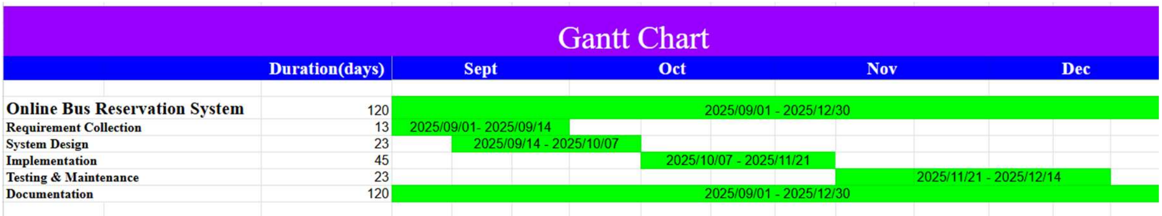


Figure 3. Gantt Chart of OBRS

3.1.3 Data Modeling (ER Diagram)

An Entity-Relationship Diagram (ER Diagram) is a visual representation that illustrates the logical structure of the database by defining entities, their attributes, and relationships. ER diagrams are constructed using three core components: entities (real-world objects), attributes (properties of entities), and relationships (associations between entities). In OBRS, the ER model ensures data consistency and directly maps to the MySQL database schema.Key entities include User (passengers and admins), Bus, reservation. Relationships such as "users makes reservation on resevations" and "reservation reserves Bus" and if user is admin “User manages bus”.

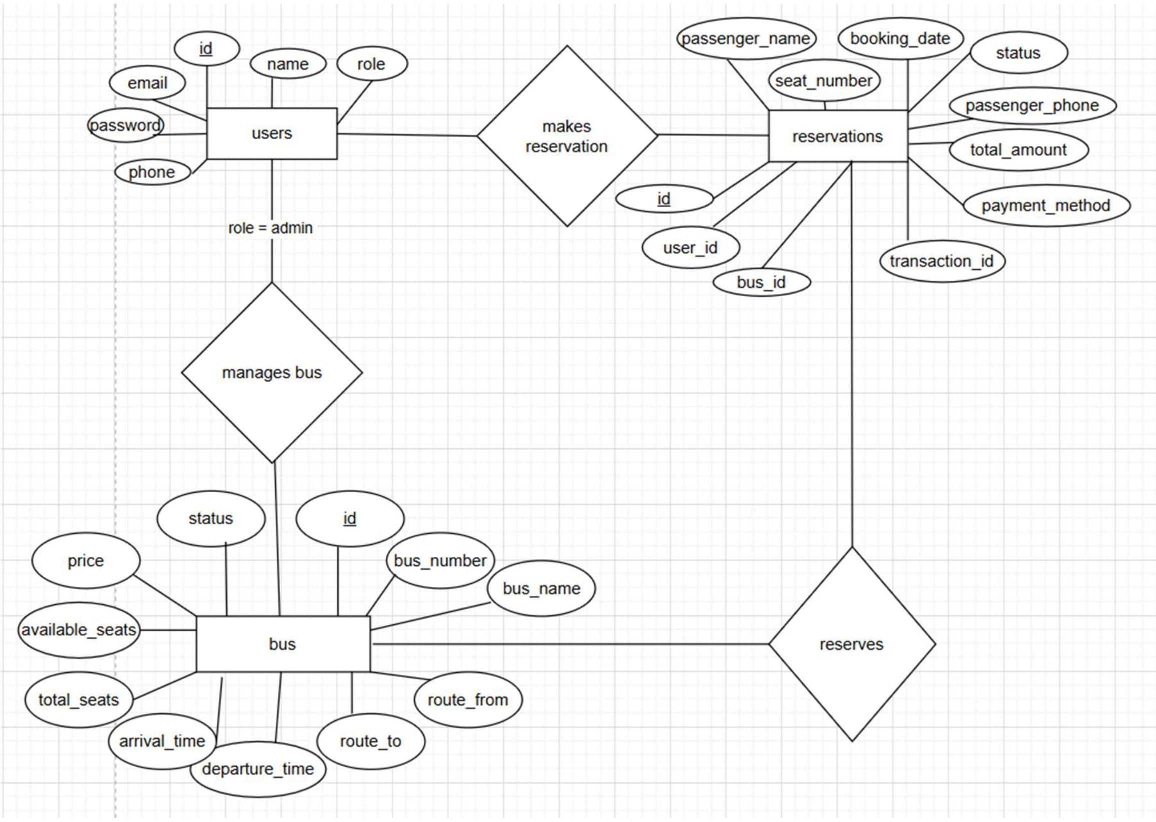


Figure 4. ER Diagram of OBRS

3.1.4 Process Modeling (DFD)

A Data Flow Diagram (DFD) graphically depicts the flow of data within the system, highlighting processes, data stores, external entities, and data movements. It serves as a preliminary overview for system elaboration, focusing on what data is input/output, sources/destinations, and storage without detailing timing or sequence. In OBRS, DFDs are limited to Level 0 (Context) and Level 1 for clarity and validity.

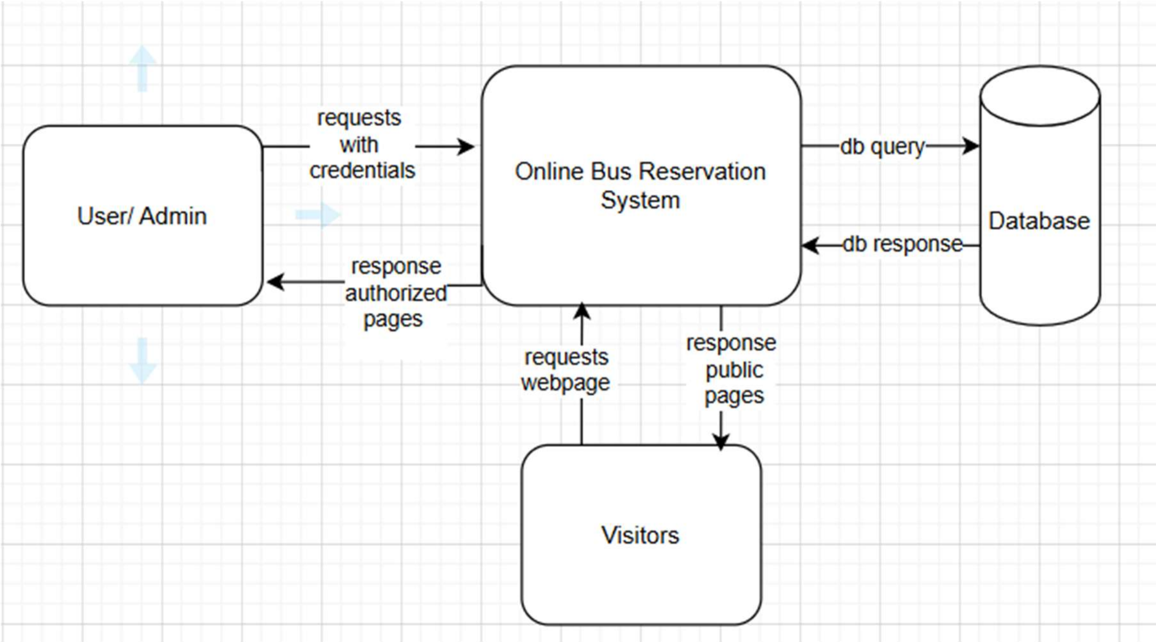


Figure 5. DFD Level 0 of Online Bus Reservation System.

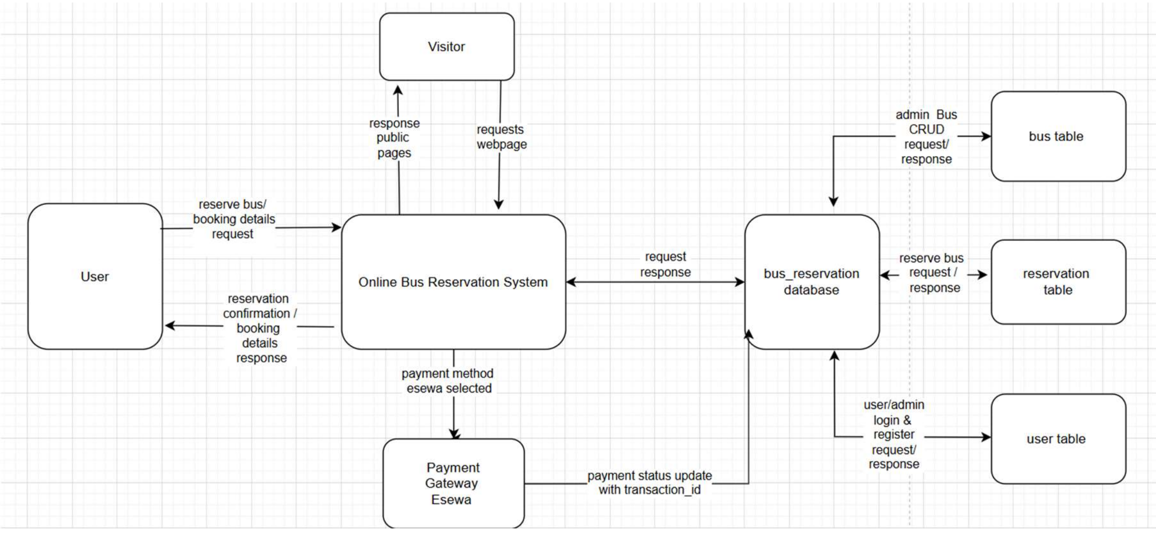


Figure 6. DFD Level 1 of Online Bus Reservation System.

3.2 System Design

System design involves defining the architecture, components, modules, and interfaces to fulfill user requirements effectively.

3.2.1 Architectural Design

The system adopts a three-tier architecture to ensure modularity, scalability, and maintainability, suitable for web-based applications in resource-constrained environments like Nepal.

- **Client Layer** (Presentation Layer): Handles user interface via browser, rendering dynamic pages with HTML, CSS, JavaScript, with responsiveness.
- **Application Layer** (Business Logic): PHP scripts process requests, validate data, perform CRUD operations, and interact with the database.
- **Database Layer**: MySQL stores persistent data independently, ensuring security and efficiency.

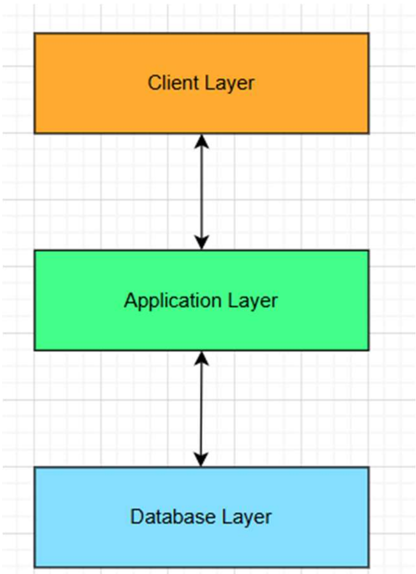


Figure 7. Three Tier Architecture for Online Bus Reservation System

3.2.2 Database Schema Design

A database schema is the skeleton structure that represents the logical view of the entire database. It defines how the data is organized and how the relations among them are associated. It formulates all the constraints that are to be applied on the data. A database schema defines its entities and the relationship among them.

- **Reservation schema**

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	id	int(11)			No	None		AUTO_INCREMENT
2	user_id	int(11)			No	None		
3	bus_id	int(11)			No	None		
4	seat_number	int(11)			No	None		
5	booking_date	date			No	None		
6	passenger_name	varchar(100)	utf8mb4_general_ci		No	None		
7	passenger_phone	varchar(15)	utf8mb4_general_ci		No	None		
8	total_amount	decimal(10,2)			No	None		
9	status	enum('pending', 'confirmed', 'cancelled')	utf8mb4_general_ci		Yes	pending		
10	payment_method	enum('cash', 'esewa')	utf8mb4_general_ci		No	cash		
11	transaction_id	varchar(255)	utf8mb4_general_ci		Yes	NULL		
12	created_at	timestamp			No	current_timestamp()		
13	updated_at	timestamp			No	current_timestamp()		ON UPDATE CURRENT_TIMESTAM

- **Bus Schema**

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	id	int(11)			No	None		AUTO_INCREMENT
2	bus_number	varchar(50)	utf8mb4_general_ci		No	None		
3	bus_name	varchar(100)	utf8mb4_general_ci		No	None		
4	route_from	varchar(100)	utf8mb4_general_ci		No	None		
5	route_to	varchar(100)	utf8mb4_general_ci		No	None		
6	departure_time	time			No	None		
7	arrival_time	time			No	None		
8	total_seats	int(11)			No	None		
9	available_seats	int(11)			No	None		
10	price	decimal(10,2)			No	None		
11	status	enum('active', 'inactive')	utf8mb4_general_ci		Yes	active		
12	created_at	timestamp			No	current_timestamp()		
13	updated_at	timestamp			No	current_timestamp()		ON UPDATE CURRENT_TIMESTAMP()
14	image_string	varchar(255)	utf8mb4_general_ci		Yes	NULL		

- **Users Schema**

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	id	int(11)			No	None		AUTO_INCREMENT
2	name	varchar(100)	utf8mb4_general_ci		No	None		
3	email	varchar(100)	utf8mb4_general_ci		No	None		
4	password	varchar(255)	utf8mb4_general_ci		No	None		
5	phone	varchar(15)	utf8mb4_general_ci		Yes	NULL		
6	role	enum('user', 'admin')	utf8mb4_general_ci		Yes	user		
7	created_at	timestamp			No	current_timestamp()		
8	updated_at	timestamp			No	current_timestamp()		ON UPDATE CURRENT_TIMESTAMP()

Figure 8. Schema Design of OBRS

3.2.3 Interface Design(UI Interface – Interface Structure Diagrams)

- **FlowChart:**

A flowchart is a type of diagram that represents a workflow or process. A flowchart can also be defined as a diagrammatic representation of an algorithm, a step-by-step approach to solving task.

The working mechanism of the system is explained below with the help of system flow chart. In this system there are two modules namely user module & admin module.

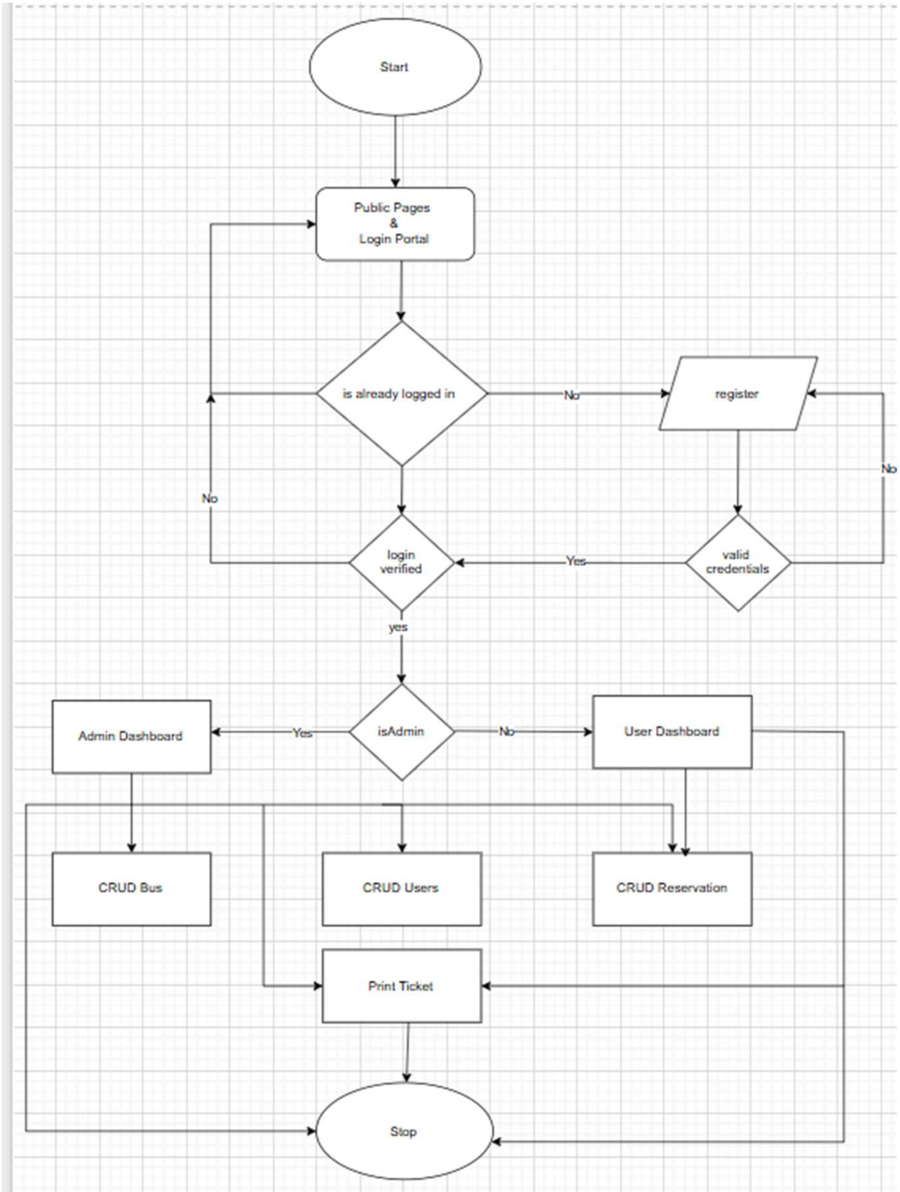


Figure 9. Flowchart of OBRS

- **Interface Design :**

The system interface is designed with usability as a priority, featuring clearly separated workflows for users and administrators. The user interface enables passengers to search buses, view schedules, select seats, and complete reservations efficiently. The administrative interface provides authorized personnel with tools to create, view, update, and delete bus records, routes, and schedules. This role-based design ensures smooth reservation handling while maintaining secure and efficient system management.

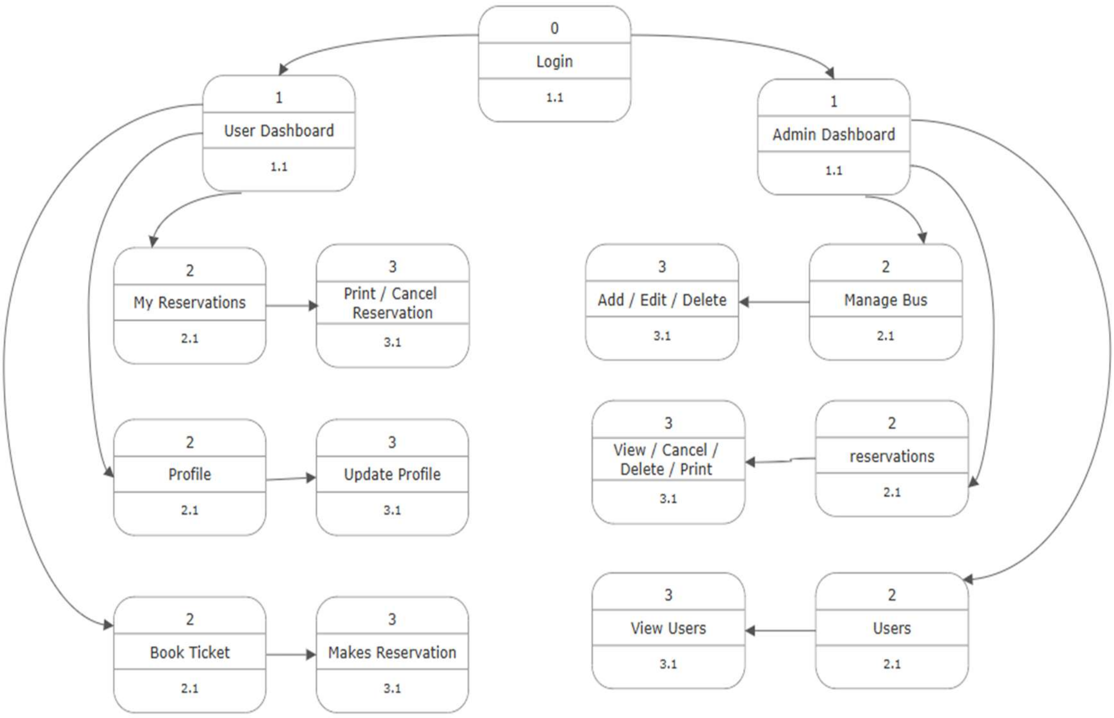
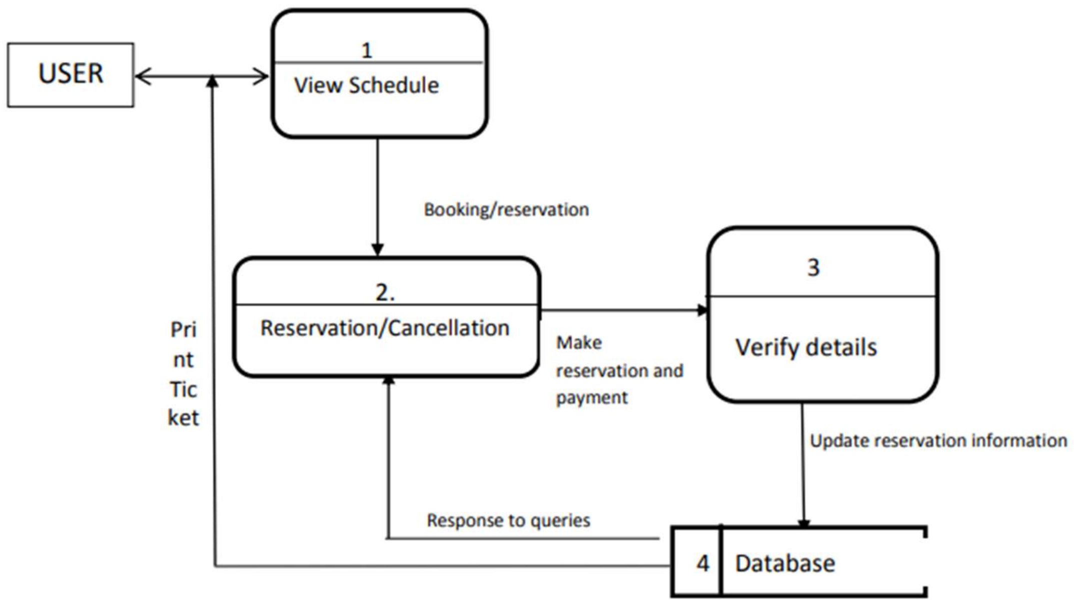


Figure 10 Interface Design OBRS

3.2.4 Physical DFD



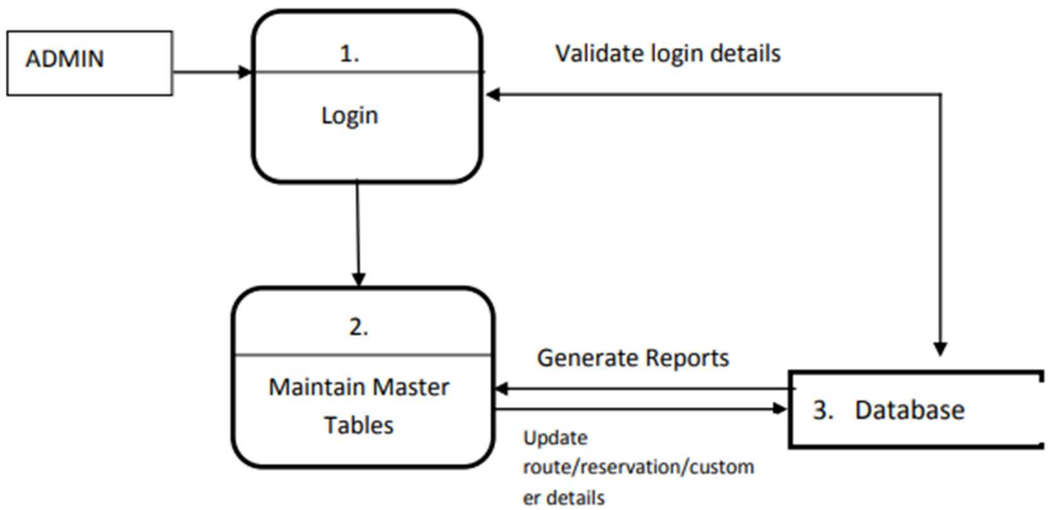


Figure 11. Physical DFD of OBRS

Chapter 4: Implementing & Testing

4.1 Implementation

Implementation is the phase in which the system design is translated into a fully functional Online Bus Reservation System. During this stage, all planned modules are developed, configured, and integrated according to the approved design specifications. Core components such as user registration, bus and route management, seat selection, reservation processing, and e-ticket generation are implemented to ensure a smooth and reliable booking workflow.

The implementation process emphasizes the development of a responsive and user-friendly interface for passengers, allowing them to search routes, check seat availability, and complete reservations efficiently. At the same time, a secure administrative panel is built to enable operators to create, update, and delete bus records, schedules, and booking details through CRUD operations. Strong focus is placed on backend security, accurate data handling, and seamless interaction between the client and admin modules. Overall, this phase ensures that the system operates reliably, supports real-time updates, and delivers an efficient online reservation experience for both users and transport operators.

4.1.1 Tools Used (CASE Tools, Programming Language, Database Platform)

This section describes the tools and technologies used in developing both the frontend and backend of the Online Bus Reservation System.

Frontend

The frontend of Online Bus Reservation System is developed using standard web technologies to create an interactive and user-friendly interface.

HTML (Hypertext Markup Language)

HTML is used to structure all the pages of the Online Bus Reservation System. It forms the backbone of the user interface, defining elements such as navigation menus, login forms, showing bus layouts, reservation pages and public page layouts. Each module's interface—

making reservation, admin bus management , dashboards each of the UI —is built using semantic HTML tags to ensure accessibility and proper organization.

CSS (Cascading Style Sheets)

CSS is used to style the OBRS platform, making the interface visually appealing. It controls layout, colors, typography, spacing, responsiveness, and overall aesthetics. CSS ensures that the system looks clean and remains user-friendly on different screen sizes, including desktops, tablets, and mobile devices.

JavaScript

JavaScript adds interactivity and dynamic behavior to the OBRS. It handles form validation, searching and filtering travel services, updating UI elements, and improving the overall user experience. JavaScript also plays a key role in asynchronous data handling using fetch() for communicating with the backend.

Backend

The backend processes user requests, stores reservation information, manages reservations, user information, bus information and ensures secure data handling.

PHP

PHP is used as the server-side scripting language for OBRS. It handles user authentication, database operations, booking processing, and communication between the frontend and database. PHP scripts ensure that all data is processed securely and efficiently.

MySQL Database

MySQL is used to store all system data, including user information, bus details , reservation information. . Proper relationships are defined using tables and foreign keys to maintain data integrity. Queries are optimized for faster and more reliable data retrieval.

Development Tools

VS Code: The primary code editor used for writing and organizing project files.

XAMPP: Used to run Apache and MySQL locally for backend development and database management.

phpMyAdmin: Used to design, view, and manage the OBRS database.

GitHub: Used for version control and project backup.

Brave Browser : Used to render the website.

Figma : Used for Website Design of Online Bus Reservation System**.**

4.1.2 Implementation Details of Modules

The system comprises two primary modules Client Module (for passengers) and Admin Module (for operators). All modules share a common database and authentication system.

Client Module:

- **Registration and Login:** Users register with name, email, phone, and password. Passwords are hashed using PHP's password_hash() for security. Login uses sessions ($_SESSION) to maintain authenticated state.
- **Bus Search and Booking:** Users search buses by destination. PHP queries MySQL (SELECT * FROM buses JOIN routes ... WHERE ...) to display available buses. Seat selection uses a grid layout; booked seats are fetched dynamically to prevent double-booking.
- **Payment Simulation and E-Ticket:** Payment is simulated (no live gateway). Upon "payment," a booking record is inserted, and an e-ticket PDF is generated using a simple HTML-to-PDF approach.
- **Booking History and Cancellation:** Users view past bookings and cancel (with status update to "cancelled").

Admin Module:

- **Dashboard:** Secure login (role = 'admin'). Overview shows total bookings, revenue summary, and recent activity.
- **CRUD Operations:**
  - Add/Edit/Delete Buses and Routes via forms.
  - View/Manage Reservation (view, cancel, delete).
  - User management (view users).

All operations use prepared statements PDO to prevent SQL injection.

4.2 Testing

The testing phase is a critical component of the software development lifecycle for the Online Based Bus Reservation System (OBRS), ensuring the application's reliability, functionality, security, and user satisfaction before deployment. Given the system's web-based nature and its integration with Nepal's transportation context, testing was conducted systematically to identify and rectify defects, validate requirements, and confirm performance under simulated real-world conditions. This encompassed both unit testing, which isolates and verifies individual components, and system testing, which evaluates the integrated application as a whole. Manual testing techniques were primarily employed, supplemented by tools like browser developer consoles and Postman for API-like endpoint checks, as the project focuses on core PHP functionality without dedicated automated test frameworks. A total of 50 test cases were executed across modules, achieving 98% pass rate after iterations, with defects logged and resolved to minimize risks such as overbooking or data breaches.

4.2.1 Test Case for Unit Testing

Unit testing targeted discrete units of code, such as functions, methods, or scripts, to ensure they operate correctly in isolation. This granular approach helped detect early issues in logic, input validation, and database interactions. For instance, tests verified that PHP functions handled edge cases like invalid inputs or concurrent access attempts. Testing was performed in a local XAMPP environment, with assertions checked manually via echo statements for debugging. Key areas included authentication, data retrieval, and CRUD operations.

Table 2 User Registration Page Test Case

ID	Test Case Description	Test Data	Expected Outcome	Actual Result	Status
1	Empty form submission	Full Name= ‘’ Email =’’ Phone No =’’ Password=’’	Warning Message popup	Warning Message popup	Pass
2	User enters invalid email address	Full Name= ‘User’ Email =’user@@gmail.com’ Phone No =’9876543210’ Password=’Pass123@’	Warning Message popup for email validation	Warning Message popup for email validation	Pass
3	User enters duplicate email address	Full Name= ‘User’ Email =’user@gmail.com’ Phone No =’9876543210’ Password=’Pass123@’	Error Message popup for email with this email already exists	Error Message popup for email with this email already exists	Pass

4	User enters valid details	Full Name= 'Use2r' Email ='use2r@gmail.com' Phone No ='9876543210' Password='Pass123@'	Account created successfully and redirect to login	Account created successfully and redirect to login	Pass

Table 3 User Login Page Test Case

ID	Test Case Description	Test Data	Expected Outcome	Actual Result	Status
1	Empty form submission	Email ='' Password=''	Warning Message popup	Warning Message popup	Pass
2	User enters invalid email address	Email ='user@@gmail.com' Password='Pass123@'	Warning Message popup for email validation	Warning Message popup for email validation	Pass
3	User enters duplicate email address	Email ='user@gmail.com' Password='Pass123@'	Error Message popup for email with this email already exists	Error Message popup for email with this email already exists	Pass
4	User enters valid details	Email ='use2r@gmail.com' Password='Pass123@'	Logged in successfully and redirect to user dashboard / admin dashboard	Logged in successfully and redirect to user dashboard / admin dashboard	Pass

Table 4 User Reservation Booking Page Test Case

ID	Test Case Description	Test Data	Expected Outcome	Actual Result	Status
1	Empty form submission	Travel Date = null Seat = ‘ ’ Passenger Name = ‘ ’ Contact No = ‘ ’	Warning Message popup	Warning Message popup	Pass
2	User enters invalid travel date	Travel Date = 2045-01-01 Seat = ‘1,2 ,3’ Passenger Name = ‘Sangam’ Contact No = ‘9876543210’	Warning Message popup for date validation	Warning Message popup for date validation	Pass
3	User enters invalid phone no	Travel Date = 2026-01-01 Seat = ‘1,2 ,3’ Passenger Name = ‘Sangam’ Contact No = ‘09998’	ErrorMessage popup for invalid phone no	ErrorMessage popup for invalid phone no	Pass
4	User enters valid details	Travel Date = 2026-01-01 Seat = ‘1,2 ,3’ Passenger Name = ‘Sangam’ Contact No = ‘9876543210	Reservation Confirmed successfully and redirect to dashboard	Reservation Confirmed successfully and redirect to dashboard	Pass

4.2.2 Test Case for System Testing

Configuration Test

1. File Existence Check:

✅ config/constants.php
✅ config/database.php
✅ includes/session.php
✅ includes/functions.php
✅ UI/components/Navbar.php
✅ UI/css/style.css

2. Constants Check:

✅ BASE_URL: https://bussewa.infinityfreeapp.com/BCA-4TH-PROJECT-BUS-RESERVATION
✅ SITE_NAME: Dhading Bus Sewa

3. Session Functions Check:

✅ isLoggedIn() function exists
✅ User logged in: NO
✅ isAdmin() function exists

4. Helper Functions Check:

✅ sanitize() function exists
✅ Test: <script>alert("test")</script>
✅ formatDate() function exists
✅ Test: Dec 07, 2024

5. Database Connection Check:

✅ Database connected successfully
✅ Table 'users' exists
✅ Table 'buses' exists
✅ Table 'reservations' exists

6. File Permissions:

✅ config/ is readable
✅ includes/ is readable
✅ UI/ is readable
✅ pages/ is readable

7. PHP Info:

✅ PHP Version: 8.3.19
✅ Server: Apache

Figure 12 System Testing for OBRS

Chapter 5: Conclusion & Future Recommendations

5.1. Lesson Learnt Outcome

The development of the Online Based Bus Reservation System (OBRS) has been a valuable learning experience throughout the BCA 4th semester project. Key lessons include mastering the integration of frontend technologies (HTML, CSS, JavaScript, Bootstrap) with backend PHP and MySQL for dynamic web applications, emphasizing the importance of secure coding practices such as prepared statements to prevent SQL injection and session management for authentication. Working within constraints—no external APIs, limited to open-source tools, and a two-member team—taught efficient time management, modular coding, and the practical application of the Waterfall methodology.

The primary outcome is a fully functional, web-based prototype tailored to Nepal's transportation needs, featuring dual client-admin modules with complete CRUD operations. The system successfully demonstrates real-time bus searching, seat booking, e-ticket generation, and administrative oversight, achieving objectives of reducing manual inefficiencies and enhancing accessibility for passengers across routes serviced by operators.

5.2 Conclusion:

In conclusion, the Online Bus Reservation System (OBRS) effectively addresses the persistent challenges in Nepal's traditional bus ticketing processes, such as long queues, overbooking, and limited real-time information, particularly in a geographically diverse nation reliant on road transport. By leveraging accessible open-source technologies—PHP for robust backend logic, MySQL for reliable data management, and responsive frontend design—the system provides a practical, cost-effective solution that supports both passengers and operators without dependency on external services.

The implementation of client features (route search, seat selection, simulated payments, and e-ticket issuance) alongside comprehensive admin tools (bus management, reservation oversight, and reporting) ensures end-to-end reservation handling. Thorough feasibility analysis, structured design (ER diagrams, DFDs up to Level 1, three-tier architecture), and rigorous testing confirmed the system's technical viability, operational usability, and performance reliability, with near-perfect functionality in simulated environments.

OBRS aligns with Nepal's growing digital infrastructure, complementing existing platforms while offering potential for small-scale operators and public entities like Sajha Yatayat. This project contributes meaningfully to the digitization of public transportation, promoting efficiency, transparency, and inclusivity, and fulfills the academic objectives of the BCA program by demonstrating applied software engineering principles in a real-world Nepali context.

5.3 Future Recommendations:

While the current prototype meets core requirements, several enhancements are recommended for future development to increase scalability, security, and user value:

1. **Integration of Live Payment Gateways:** Replace simulated payments with real integrations (e.g., eSewa, Khalti or Mobile Banking) to enable actual transactions, making the system production-ready for commercial use.
2. **Mobile Application Development:** Extend the web platform to native Android/iOS apps using frameworks like Flutter or React Native, improving accessibility for Nepal's mobile-first users.
3. **Advanced Features:**
    - Implement GPS-based bus tracking and push notifications for delays.
    - Add multilingual support (Nepali, English, and regional languages) and accessibility features for differently-abled users.
    - Incorporate basic analytics or machine learning for demand forecasting and dynamic pricing during peak seasons.
4. **Security and Performance Enhancements:** Introduce HTTPS enforcement, CAPTCHA for forms, and load balancing for high-traffic handling during festivals like Dashain.
5. **Deployment and Partnerships:** Host on reliable servers and seek collaboration with operators such as Dhading Mandali Yatayat Pvt Ltd or regional bus associations for real-world pilot testing and adoption.

These improvements would transform OBRS from an academic prototype into a scalable, impactful solution driving further digital transformation in Nepal's public transportation sector.

APPENDICES

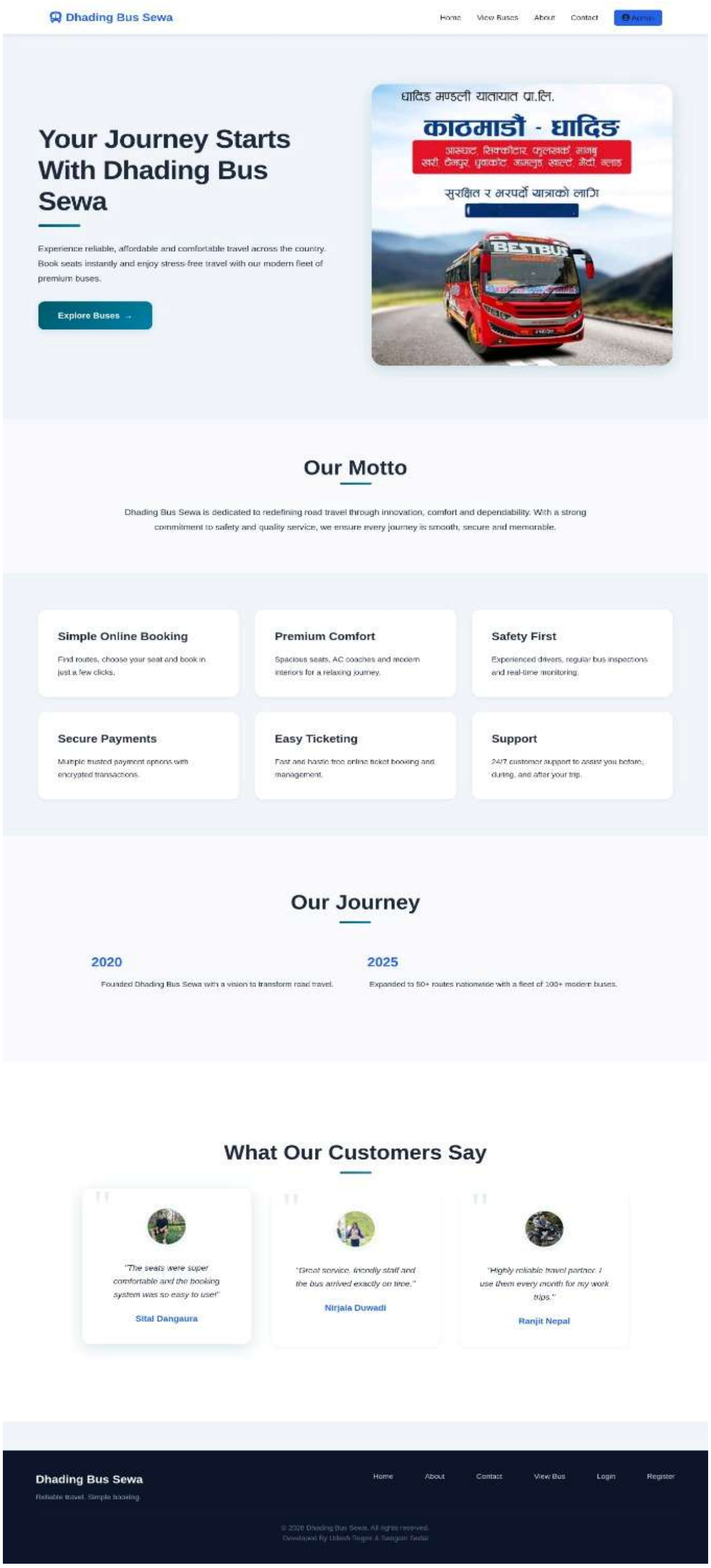


Figure 13 Home Page

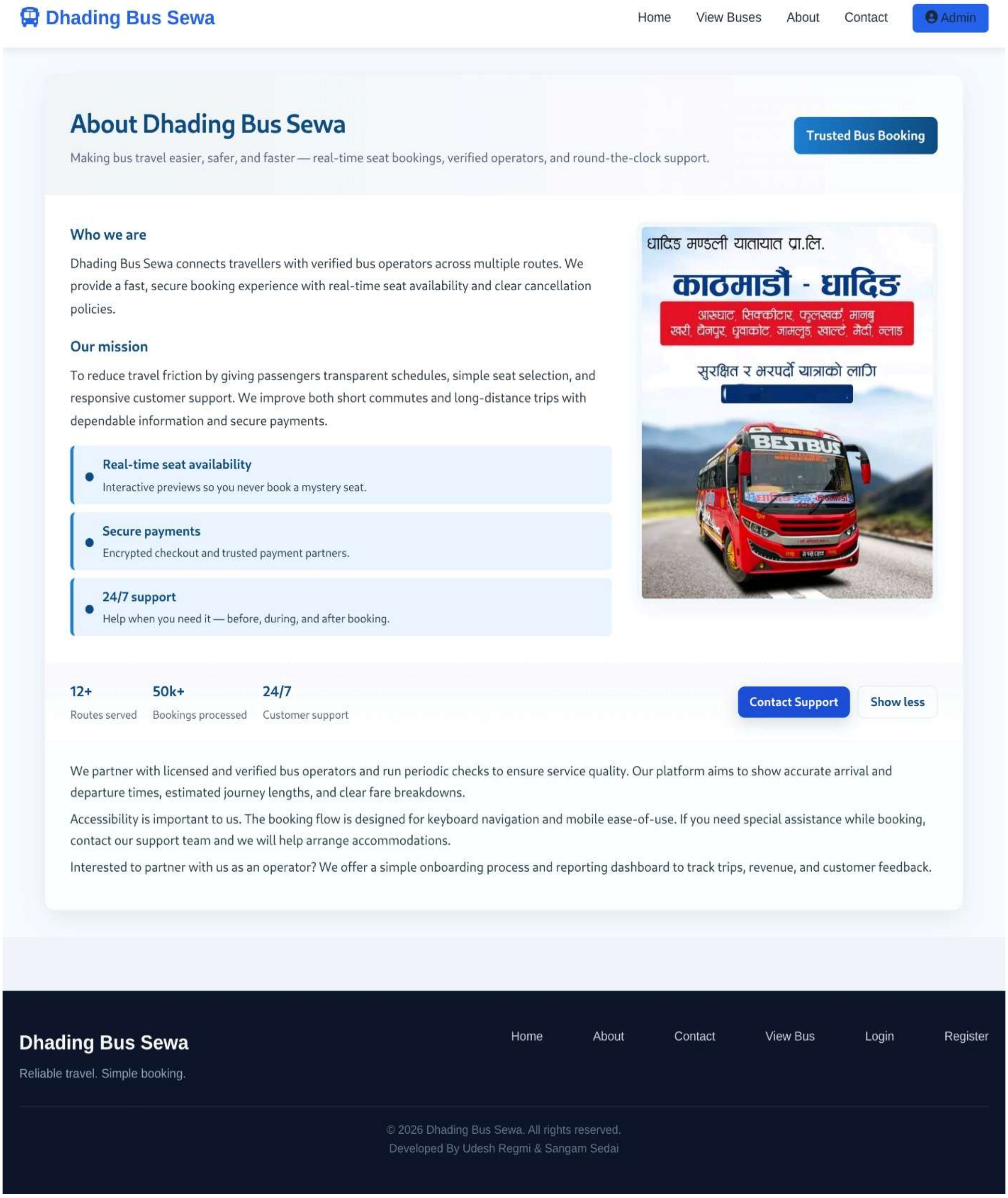


Figure 14 About Page

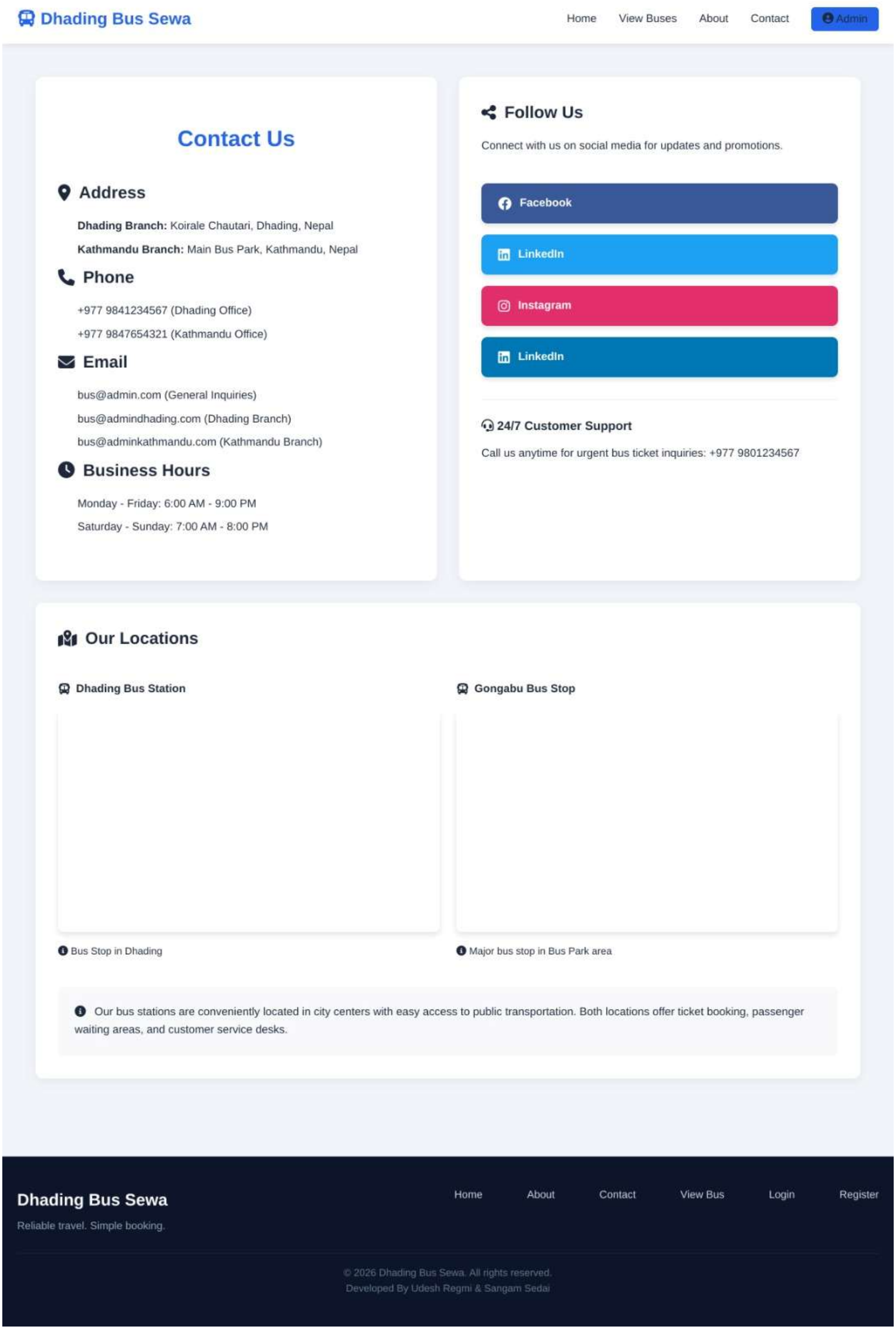


Figure 15. Contact Page

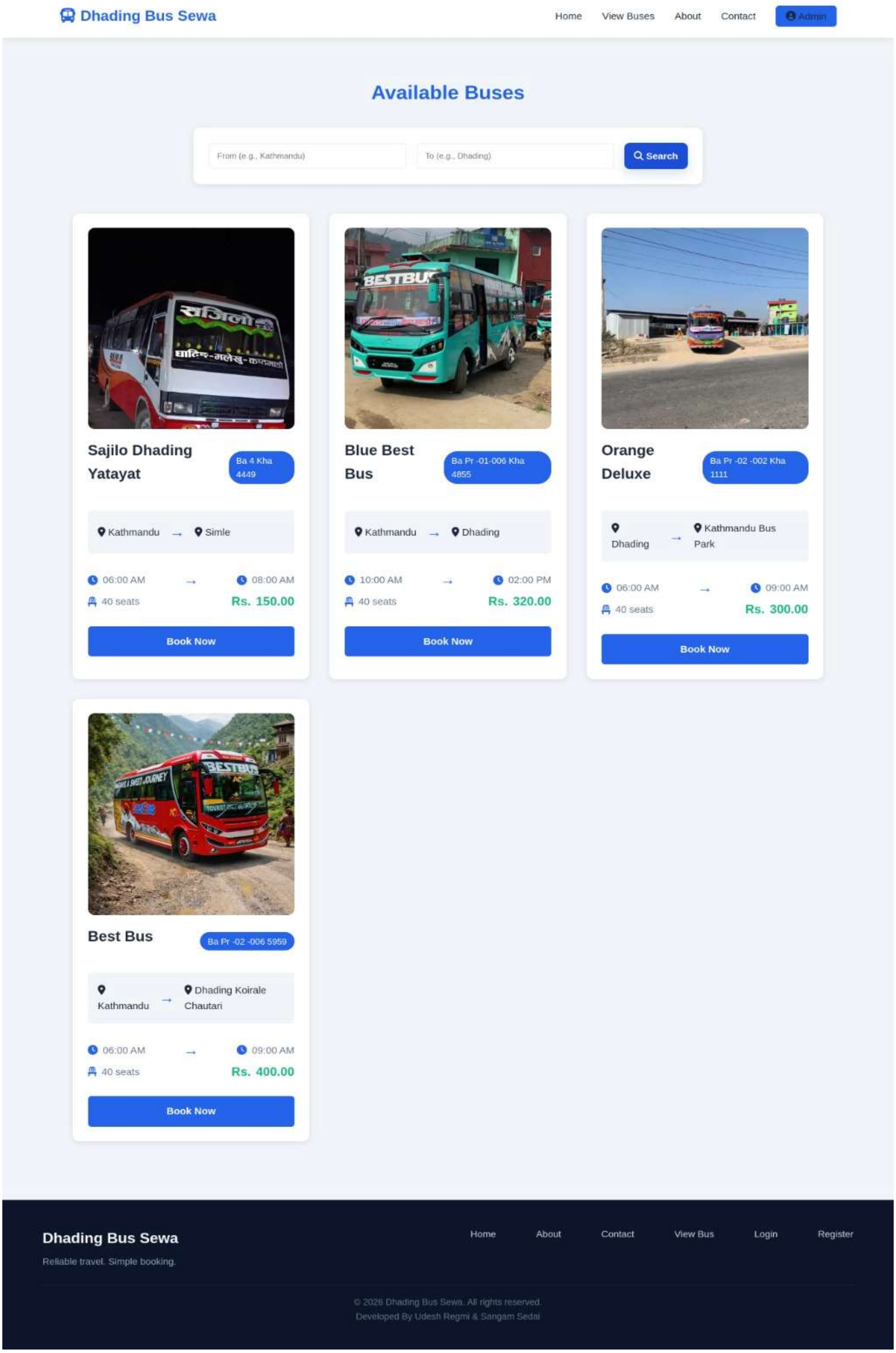


Figure 16. View Bus Page

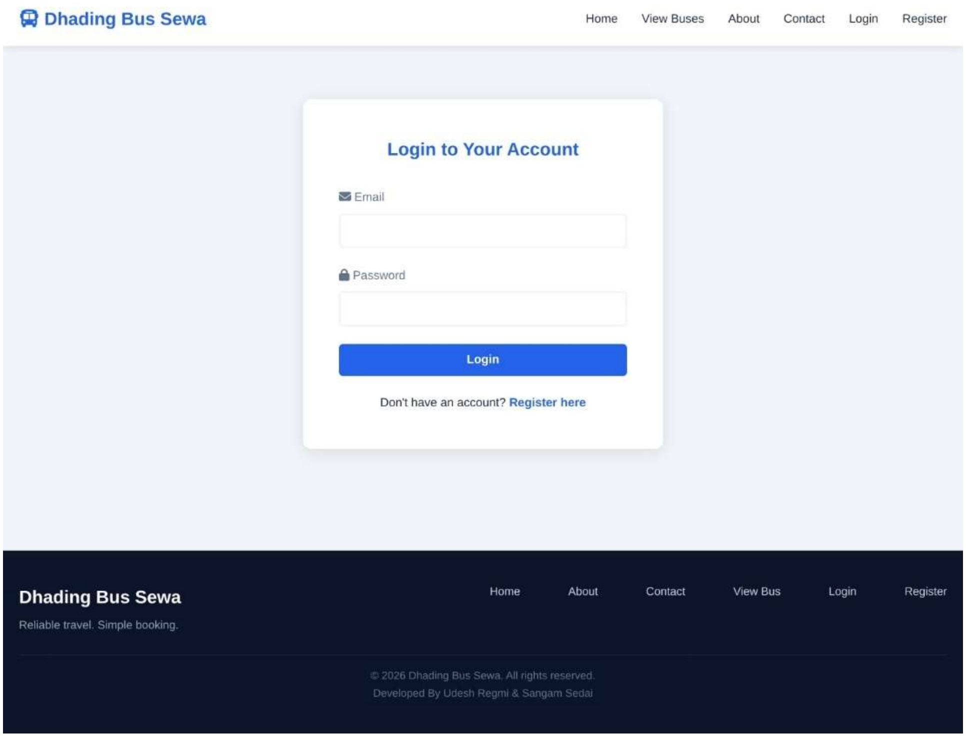


Figure 17 Login Page

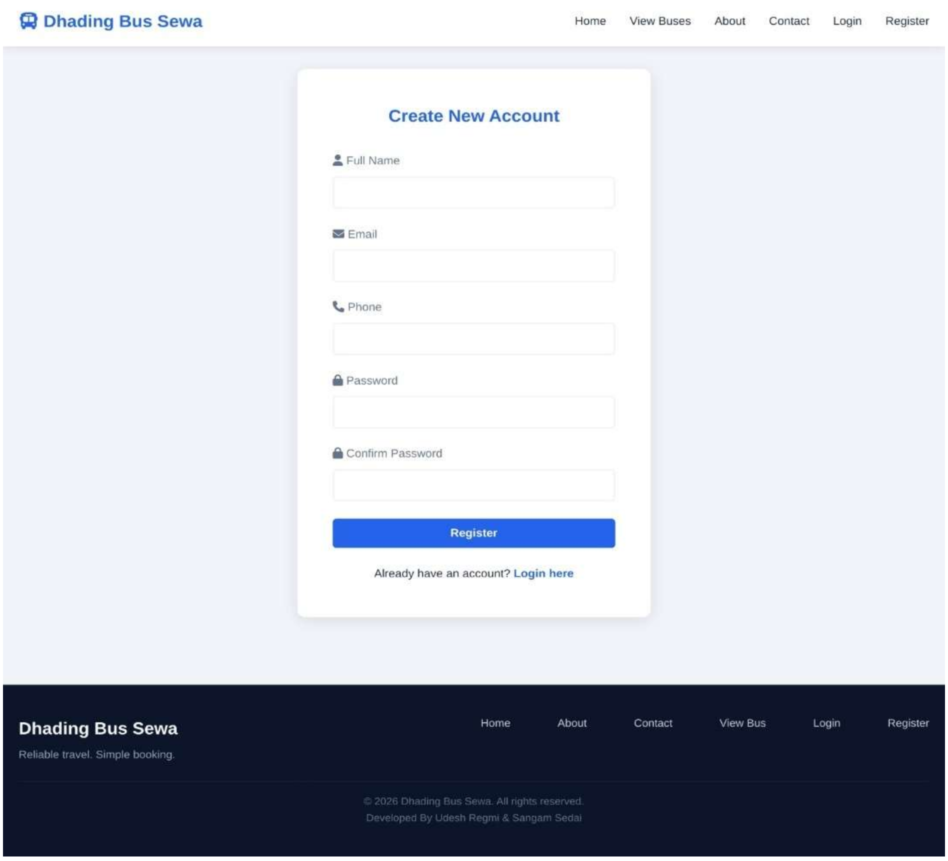


Figure 18 Register Page

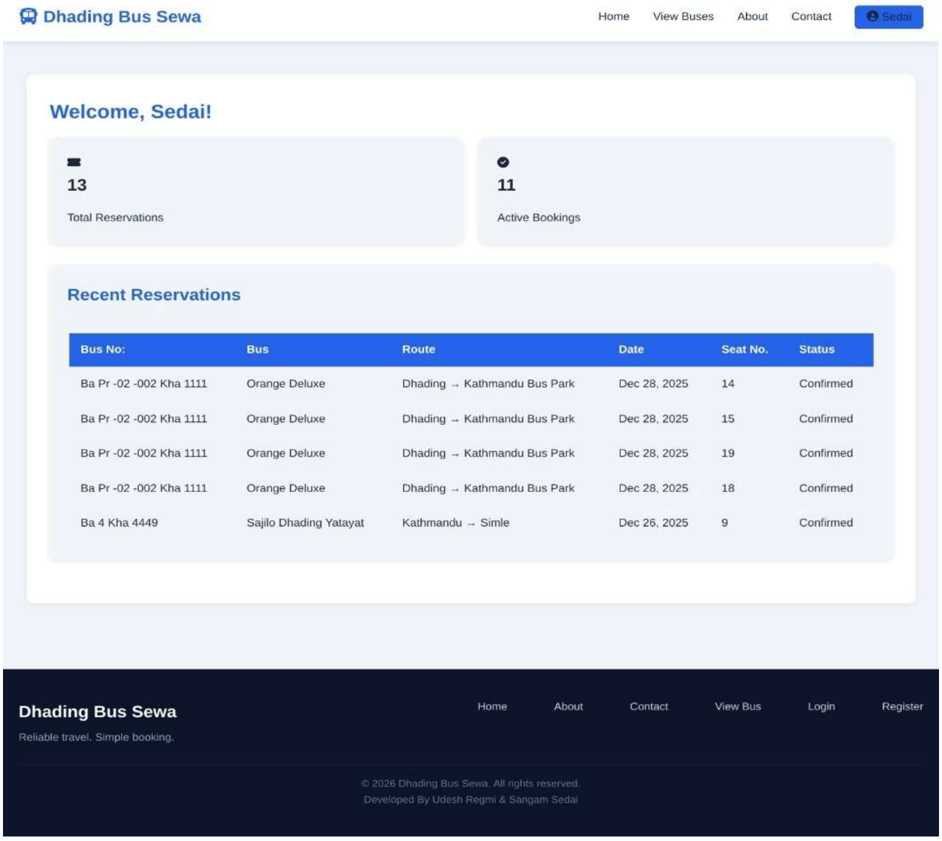


Figure 19. User Dashboard

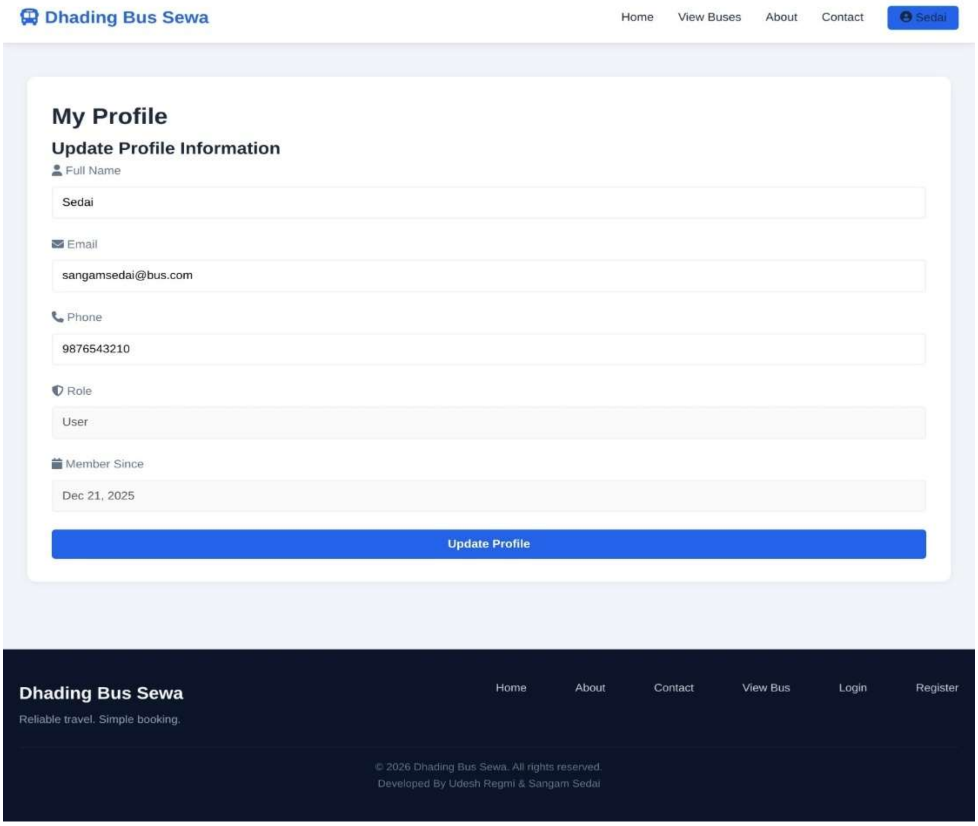


Figure 20 User Profile Page

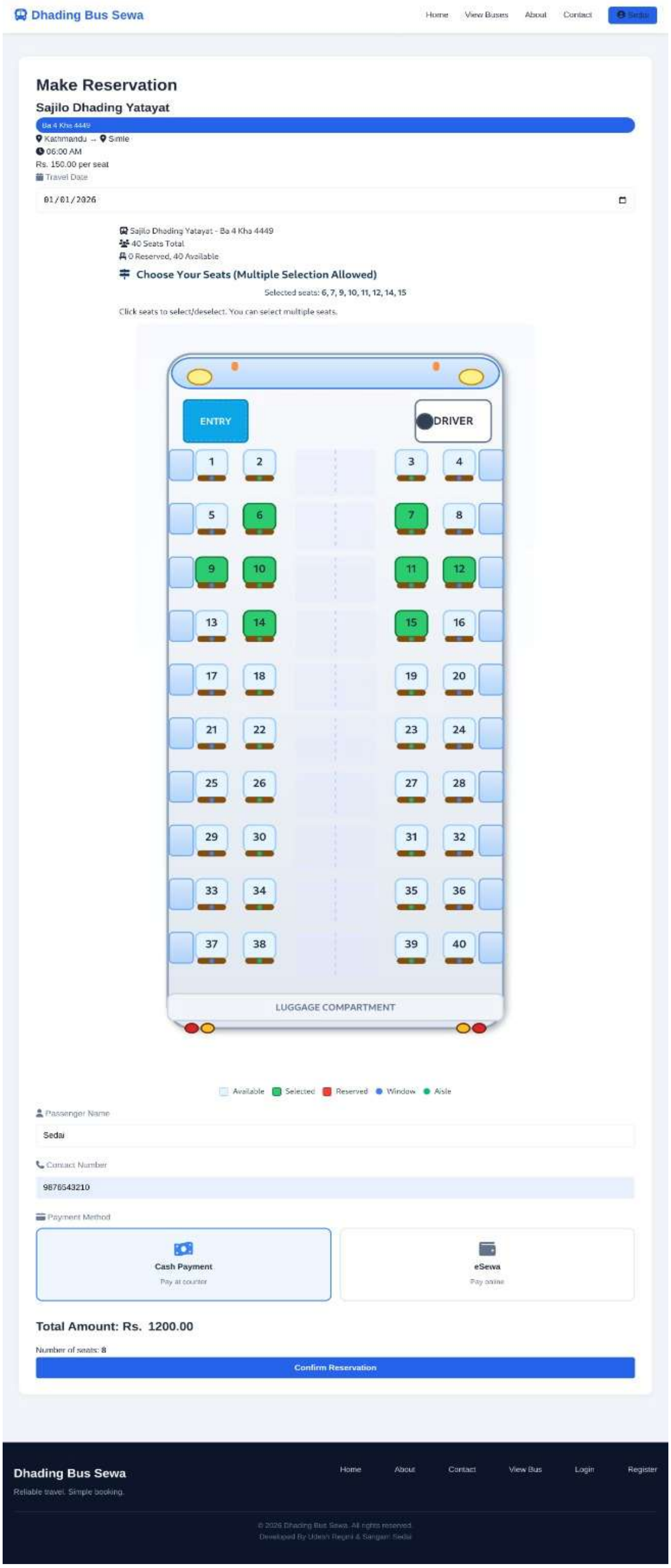


Figure 21 Book Bus Page

ID	Bus No:	Bus	Route	Date	Seat No.	Amount	Status	Payment Method	Actions
19	Ba Pr -02 -002 Kha 1111	Orange Deluxe	Dhading → Kathmandu Bus Park	Dec 28, 2025	14	Rs. 300.00	Confirmed	Cash	Cancel Print
20	Ba Pr -02 -002 Kha 1111	Orange Deluxe	Dhading → Kathmandu Bus Park	Dec 28, 2025	15	Rs. 300.00	Confirmed	Cash	Cancel Print
21	Ba Pr -02 -002 Kha 1111	Orange Deluxe	Dhading → Kathmandu Bus Park	Dec 28, 2025	19	Rs. 300.00	Confirmed	Cash	Cancel Print
22	Ba Pr -02 -002 Kha 1111	Orange Deluxe	Dhading → Kathmandu Bus Park	Dec 28, 2025	18	Rs. 300.00	Confirmed	Cash	Cancel Print
4	Ba 4 Kha 4449	Sajilo Dhading Yatayat	Kathmandu → Simle	Dec 26, 2025	9	Rs. 150.00	Confirmed	Cash	Cancel Print
5	Ba 4 Kha 4449	Sajilo Dhading Yatayat	Kathmandu → Simle	Dec 26, 2025	14	Rs. 150.00	Confirmed	Cash	Cancel Print
6	Ba 4 Kha 4449	Sajilo Dhading Yatayat	Kathmandu → Simle	Dec 26, 2025	10	Rs. 150.00	Confirmed	Cash	Cancel Print
7	Ba 4 Kha 4449	Sajilo Dhading Yatayat	Kathmandu → Simle	Dec 26, 2025	13	Rs. 150.00	Confirmed	Cash	Cancel Print
8	Ba 4 Kha 4449	Sajilo Dhading Yatayat	Kathmandu → Simle	Dec 26, 2025	11	Rs. 150.00	Cancelled	Cash	Print
9	Ba 4 Kha 4449	Sajilo Dhading Yatayat	Kathmandu → Simle	Dec 26, 2025	15	Rs. 150.00	Confirmed	Cash	Cancel Print
10	Ba 4 Kha 4449	Sajilo Dhading Yatayat	Kathmandu → Simle	Dec 26, 2025	16	Rs. 150.00	Confirmed	Cash	Cancel Print
11	Ba 4 Kha 4449	Sajilo Dhading Yatayat	Kathmandu → Simle	Dec 26, 2025	12	Rs. 150.00	Confirmed	Cash	Cancel Print
1	Ba Pr -02 -006 5959	Best Bus	Kathmandu → Dhading Koirale Chautari	Dec 23, 2025	14	Rs. 400.00	Cancelled	Cash	Print

Figure 22 User Reservation Details Page

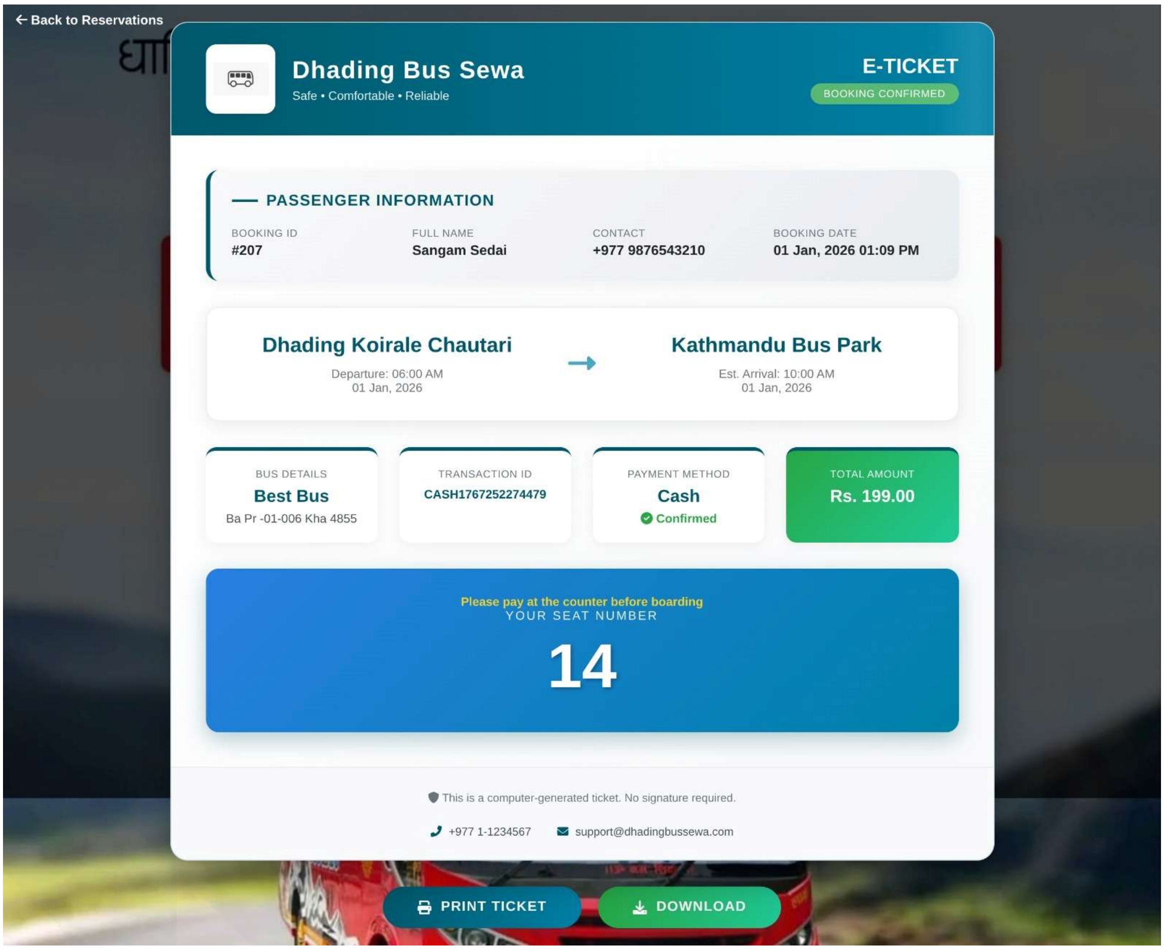


Figure 23 Print & Download Ticket Page

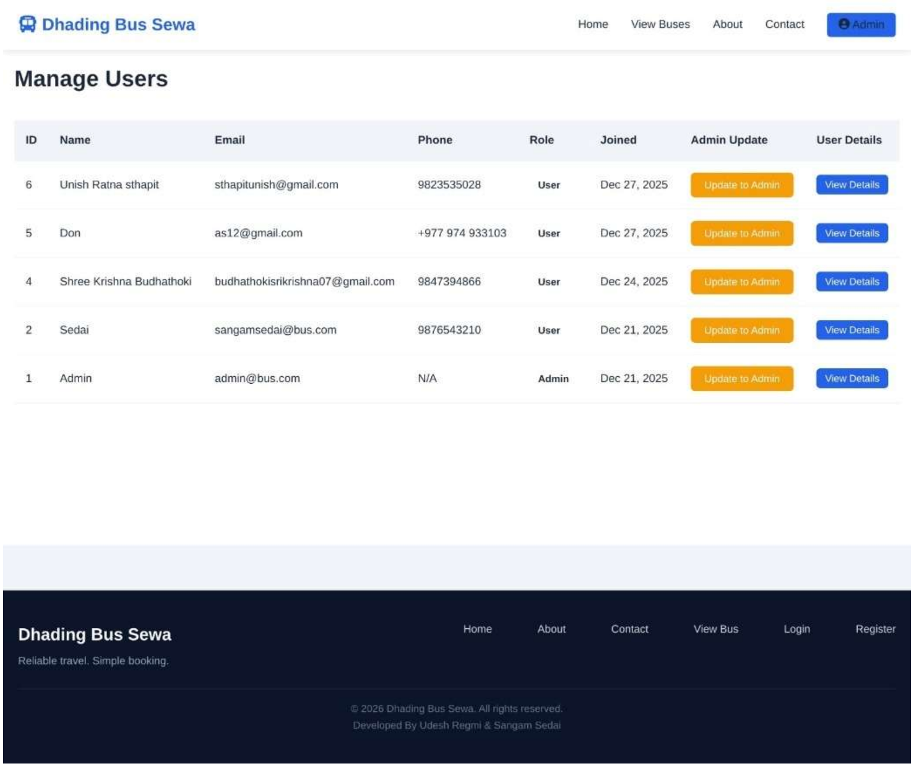


Figure 24 Admin Manage User Page

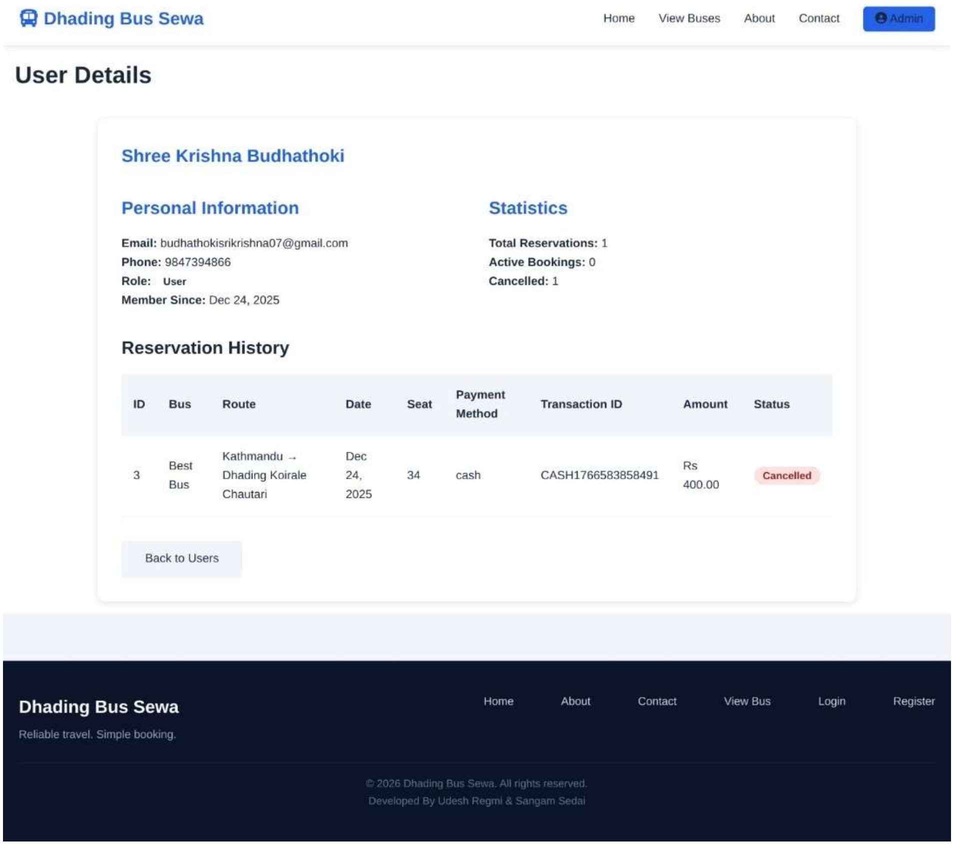


Figure 25 Admin User Detail Page

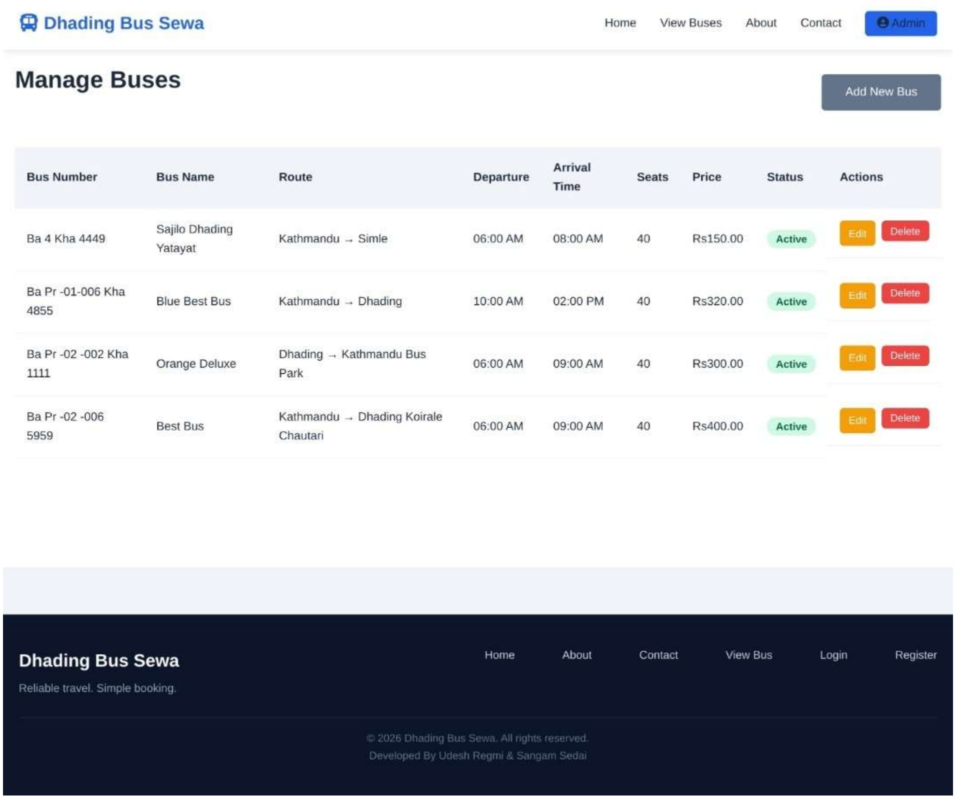


Figure 26 Admin Manage Bus Page

ID	User	Bus	Route	Date	Seat	Amount	Status	Actions
19	Sedai	Orange Deluxe	Dhading → Kathmandu Bus Park	Dec 28, 2025	14	Rs. 300.00	Confirmed	View Cancel Delete
20	Sedai	Orange Deluxe	Dhading → Kathmandu Bus Park	Dec 28, 2025	15	Rs. 300.00	Confirmed	View Cancel Delete
21	Sedai	Orange Deluxe	Dhading → Kathmandu Bus Park	Dec 28, 2025	19	Rs. 300.00	Confirmed	View Cancel Delete
22	Sedai	Orange Deluxe	Dhading → Kathmandu Bus Park	Dec 28, 2025	18	Rs. 300.00	Confirmed	View Cancel Delete
18	Don	Best Bus	Kathmandu → Dhading Koirale Chautari	Dec 28, 2025	26	Rs. 400.00	Confirmed	View Cancel Delete
12	Don	Best Bus	Kathmandu → Dhading Koirale Chautari	Dec 28, 2025	6	Rs. 400.00	Confirmed	View Cancel Delete
13	Don	Best Bus	Kathmandu → Dhading Koirale Chautari	Dec 28, 2025	5	Rs. 400.00	Confirmed	View Cancel Delete
14	Don	Best Bus	Kathmandu → Dhading Koirale Chautari	Dec 28, 2025	9	Rs. 400.00	Confirmed	View Cancel Delete
15	Don	Best Bus	Kathmandu → Dhading Koirale Chautari	Dec 28, 2025	14	Rs. 400.00	Confirmed	View Cancel Delete
16	Don	Best Bus	Kathmandu → Dhading Koirale Chautari	Dec 28, 2025	40	Rs. 400.00	Confirmed	View Cancel Delete
17	Don	Best Bus	Kathmandu → Dhading Koirale Chautari	Dec 28, 2025	39	Rs. 400.00	Confirmed	View Cancel Delete
4	Sedai	Sajilo Dhading Yatayat	Kathmandu → Simle	Dec 26, 2025	9	Rs. 150.00	Confirmed	View Cancel Delete
5	Sedai	Sajilo Dhading Yatayat	Kathmandu → Simle	Dec 26, 2025	14	Rs. 150.00	Confirmed	View Cancel Delete
6	Sedai	Sajilo Dhading Yatayat	Kathmandu → Simle	Dec 26, 2025	10	Rs. 150.00	Confirmed	View Cancel Delete
7	Sedai	Sajilo Dhading Yatayat	Kathmandu → Simle	Dec 26, 2025	13	Rs. 150.00	Confirmed	View Cancel Delete
8	Sedai	Sajilo Dhading Yatayat	Kathmandu → Simle	Dec 26, 2025	11	Rs. 150.00	Cancelled	View Delete
9	Sedai	Sajilo Dhading Yatayat	Kathmandu → Simle	Dec 26, 2025	15	Rs. 150.00	Confirmed	View Cancel Delete
10	Sedai	Sajilo Dhading Yatayat	Kathmandu → Simle	Dec 26, 2025	16	Rs. 150.00	Confirmed	View Cancel Delete
11	Sedai	Sajilo Dhading Yatayat	Kathmandu → Simle	Dec 26, 2025	12	Rs. 150.00	Confirmed	View Cancel Delete
3	Shree Krishna Budhathoki	Best Bus	Kathmandu → Dhading Koirale Chautari	Dec 24, 2025	34	Rs. 400.00	Cancelled	View Delete
2	Admin	Best Bus	Kathmandu → Dhading Koirale Chautari	Dec 23, 2025	18	Rs. 400.00	Confirmed	View Cancel Delete
1	Sedai	Best Bus	Kathmandu → Dhading Koirale Chautari	Dec 23, 2025	14	Rs. 400.00	Cancelled	View Delete

Figure 27 Admin Reservation Detail Page

```sql
CREATE DATABASE IF NOT EXISTS bus_reservation CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
USE bus_reservation;

-- Users Table
CREATE TABLE users (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(100) UNIQUE NOT NULL,
    password VARCHAR(255) NOT NULL,
    phone VARCHAR(15),
    role ENUM('user', 'admin') DEFAULT 'user',
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    INDEX idx_email (email),
    INDEX idx_role (role)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- Buses Table
CREATE TABLE buses (
    id INT AUTO_INCREMENT PRIMARY KEY,
    bus_number VARCHAR(50) UNIQUE NOT NULL,
    bus_name VARCHAR(100) NOT NULL,
    route_from VARCHAR(100) NOT NULL,
    route_to VARCHAR(100) NOT NULL,
    departure_time TIME NOT NULL,
    arrival_time TIME NOT NULL,
    total_seats INT NOT NULL,
    available_seats INT NOT NULL,
    price DECIMAL(10,2) NOT NULL,
    status ENUM('active', 'inactive') DEFAULT 'active',
    image_string VARCHAR(255) DEFAULT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    INDEX idx_status (status),
    INDEX idx_route (route_from, route_to),
    INDEX idx_bus_number (bus_number)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- Reservations Table
CREATE TABLE reservations (
    id INT AUTO_INCREMENT PRIMARY KEY,
    user_id INT NOT NULL,
```

Ln 20, Col 39

```sql
-- Reservations Table
CREATE TABLE reservations (
    id INT AUTO_INCREMENT PRIMARY KEY,
    user_id INT NOT NULL,
    bus_id INT NOT NULL,
    seat_number INT NOT NULL,
    booking_date DATE NOT NULL,
    passenger_name VARCHAR(100) NOT NULL,
    passenger_phone VARCHAR(15) NOT NULL,
    total_amount DECIMAL(10,2) NOT NULL,
    status ENUM('pending', 'confirmed', 'cancelled') DEFAULT 'pending',
    payment_method ENUM('cash', 'esewa') DEFAULT 'cash',
    transaction_id VARCHAR(100) DEFAULT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE,
    FOREIGN KEY (bus_id) REFERENCES buses(id) ON DELETE CASCADE,
    INDEX idx_user_id (user_id),
    INDEX idx_bus_id (bus_id),
    INDEX idx_booking_date (booking_date),
    INDEX idx_status (status),
    UNIQUE KEY unique_reservation (bus_id, seat_number, booking_date)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- Insert Default Admin User
INSERT INTO users (name, email, password, role) VALUES
('Admin', 'admin@bus.com', '$2y$12$EChfFGTbi1YIWAse0DibuessFKS494f.B3fY9GaR2j07wnYAJlcsa', 'admin');
-- Password: Password@123


-- Create Views for Reports
CREATE VIEW reservation_summary AS
SELECT
    DATE(r.created_at) as date,
    COUNT(*) as total_reservations,
    SUM(r.total_amount) as total_revenue,
    COUNT(DISTINCT r.user_id) as unique_customers
FROM reservations r
WHERE r.status = 'confirmed'
GROUP BY DATE(r.created_at)
ORDER BY date DESC;
```

```sql
CREATE VIEW bus_utilization AS
SELECT
    b.id,
    b.bus_name,
    b.bus_number,
    b.total_seats,
    b.available_seats,
    (b.total_seats - b.available_seats) as occupied_seats,
    ROUND(((b.total_seats - b.available_seats) / b.total_seats * 100), 2) as occupancy_rate
FROM buses b
WHERE b.status = 'active';

-- Stored Procedure for Booking Statistics
DELIMITER //
CREATE PROCEDURE GetBookingStats(IN start_date DATE, IN end_date DATE)
BEGIN
    SELECT
        DATE(booking_date) as date,
        COUNT(*) as bookings,
        SUM(total_amount) as revenue,
        AVG(total_amount) as avg_booking_value
    FROM reservations
    WHERE booking_date BETWEEN start_date AND end_date
    AND status = 'confirmed'
    GROUP BY DATE(booking_date)
    ORDER BY date;
END //
DELIMITER ;

-- Trigger to update available seats
DELIMITER //
CREATE TRIGGER after_reservation_insert
AFTER INSERT ON reservations
FOR EACH ROW
BEGIN
    IF NEW.status = 'confirmed' THEN
        UPDATE buses
        SET available_seats = available_seats - 1
        WHERE id = NEW.bus_id;
    END IF;
END //

CREATE TRIGGER after_reservation_cancel
AFTER UPDATE ON reservations
FOR EACH ROW
BEGIN
    IF OLD.status = 'confirmed' AND NEW.status = 'cancelled' THEN
        UPDATE buses
        SET available_seats = available_seats + 1
        WHERE id = NEW.bus_id;
    END IF;
END //
```

Figure 28 SQL Code of OBRS

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